Vowel Hiatus Resolution in Setswana

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Abstract

This study provides a comprehensive description of vowel hiatus resolution (VHR) in Setswana, an agglutinative S31 Bantu language, spoken in Botswana, South Africa, Zimbabwe, and Namibia (Bennett et al., 2016). Setswana has a CV/V syllable structure. Consecutive vowel sequences, known as vowel hiatus, can be found across syllable, morpheme, and word boundaries (Casali, 2011). Setswana has a syllable structure constraint disallows the occurrence of any long CV-V vowels, and therefore prevents vowel hiatus sequences (Kadenge and Simango, 2014). Consequently, various vowel hiatus resolution strategies such as vowel elision, glide formation, diphthongization, and glottal stop epenthesis are employed to repair such sequences, resulting in re-syllabification of hetero-syllabic consecutive vowels sequences into single CV syllables, or adjacent CV.CV syllables. Phonetic data collected from a Setswana speaker is used to analyse the different VHR strategies employed at various morpho-phonological contexts, both word-internally at 1) Morpheme boundaries, and 2) Syllable boundaries, and also across 3) Word boundaries. This study demonstrates that the type of hiatus resolution is dependent on two factors; the features of the vowels in combination, and the morpho-phonological contexts the hiatus sequences occur in. In other words, different morpho-phonological boundaries license or block different VHR strategies, resulting in variations in each context. At morpheme boundaries, hiatus sequences consisting of high and close-mid V1s have a tendency to glide when followed by a lower V2, whereas close-mid and open-mid V1s undergo elision when followed by higher V2. By contrast, vowel sequences resolved using V1 elision at morpheme boundaries are resolved using diphthongization at syllable boundaries. VHR across word-boundaries is always resolved using glottal stop epenthesis regardless of the vowel combination.

1 Introduction

Vowel hiatus refers to the occurrence of consecutive vowel sequences in separate syllables (Casali, 2011). Hiatus sequences can be found at various morpho-phonological contexts such as morpheme boundaries, syllable boundaries, and word boundaries. Languages vary in their treatment of hiatus sequences (Casali, 2021); they either allow it freely, allow it only in certain contexts, or disallow it completely (Kadenge and Simango, 2014). Languages that place constraints on vowel hiatus sequences use various strategies, known as vowel hiatus resolution (VHR), to eliminate them where they appear underlyingly (Casali, 2021). Some commonly used VHR strategies include vowel elision, glide formation, coalescence, diphthongization, and epenthesis (Casali, 2021). The nature of the hiatus resolution strategy employed can differ based on the morpho-phonological context and the features of the vowels involved in the hiatus sequence (Kadenge and Simango, 2014; Vratsanos and Kadenge, 2017). Previous

studies on vowel hiatus resolution have been conducted on many languages spanning across different language families, including Ngiti (Kutsch Lojenga, 1994), Indonesian (Rosenthall, 1997), Obolo (Faraclas, 1982), Eastern Ojibwa (Howard, 1972), Kavalan (Lin, 2018), Korean (De Haas, 2010), and Greek (Baltazani, 2006). Major research on the general principles of vowel hiatus resolution that apply cross-linguistically has been done by Roderic F. Casali (1996; 1997; 2011; 2021), and provides a useful guideline for analysis of VHR in Setswana. This study contributes to the general typology of VHR in languages across the world.

Setswana (ISO 639-3: tsn), also known as Sechuana or Tswana, is the national language of the Republic of Botswana (Motzafi-Haller et al., 2006). It is also the official language of Botswana, alongside English (Tsonope, 1988). It is also widely spoken in South Africa, Zimbabwe and Namibia (Bennett et al., 2016), with an estimated speaker population of 13,745,730 (Lewis, 2009) across all countries. Setswana is an S31 Bantu language, (Batibo, 2016) classified among the Sotho-Tswana (S30) group within the South-eastern zone of Bantu languages (Cole, 1955; Van de Velde et al., 2019). Typical of Bantu languages, it has highly agglutinative morphology (Tsonope, 1988). Setswana's syllables generally follow a CV or V structure (Bennett et al., 2016). Crucially, the language allows V-initial stems, and its usually mono-syllabic affixes can have both CV or V vowel-final structures. Consequently, morpho-syntactic concatenation can result in vowel hiatus (CV-V) sequences both word internally at morpheme and syllable boundaries, and across word boundaries (Kadenge and Simango, 2014). An example of vowel hiatus at prefix-stem morpheme boundaries in nouns is given below.

(1) *m^{w}ìt^{h}ùdì*

[1] *m**ò-ì**t^{h}ùdì*
1-student
'student'

Setswana generally disallows such hiatus sequences and utilizes various VHR strategies such as vowel elision, glide formation, diphthongization, and glottal stop epenthesis. Interestingly, VHR is not merely a segmental phenomenon. Setswana is a tonal language with a privative tone system, with contrasts between marked (H)igh tone and inert (L)ow tone segments (Bennett et al., 2016; Chebanne et al., 1997; Hyman, 2001). Upon the application of VHR, the tones on the vowels affected by VHR undergo deletion. While descriptions of Setswana's phonological systems and processes have previously appeared in the literature (Bennett et al., 2016; Chebanne et al., 1997; Cole, 1955; Krüger and Pretorius, 2006), VHR is a robust process in Setswana that has yet to be studied in depth.

[1] The glossing convention used throughout the paper is as follows:
The first line contains the surface representation transcribed in IPA; this line depicts the outcome of VHR on the underlying structure. The second line represents the underlying morpheme structure, which is separated by hyphens. The corresponding gloss is provided in the third line, followed by the English translation in the fourth line.

Evidently, VHR is an active process that affects consecutive vowel sequences in Setswana, necessitating systematic and careful analysis. To that end, this study provides an acoustic analysis and description of VHR in Setswana. The analysis is based on phonetic data collected from a native Bostwanan speaker and contributes information on VHR to the phonetics and phonology of the language. It provides an analysis of VHR at three morpho-phonological contexts, namely

1. Morpheme-boundaries, specifically prefix-stem boundaries in nominal *noun class-noun stem* (2-a) and verbal *subject agreement marker-verb stem* (2-b) constructions.

(2) a. *sìpònè*

sè-ìpònè
7-mirror
'mirror'

b. *báɔpélá dì-pìnà*

bá-ɔ̀pèlà dì-pìnà
2PL-sing.PRS-FV 8-song
'They sing songs.'

2. Syllable-boundaries, between hetero-syllabic adjacent vowels (3-a) that are not morpheme-boundaries

(3) a. *pòunàmà*

pòùnàmà
9.lip
'lip'

3. Word-boundaries between noun-noun (4-b) and verb-noun (4-a) constructions.

(4) a. *kè-bɔ̀n-à$^{?}$ɔ̀vɛ̀nɛ̀*

kè-bɔ̀nà ɔ̀vɛ̀nɛ̀
1SG-see.PRS-FV 9.oven
'I see an oven.'

b. *rè-fá mòsàdì$^{?}$èmɛ̀rɛ̀*

rè-f-á mò-sàdì èmɛ̀rɛ̀
1PL-give.PRS-FV 1-woman 9.bucket
'We give the woman a bucket.'

This study demonstrates that, in Setswana, hiatus resolution is driven by a syllable structure constraint that disallows the occurrence of long vowels, and hence, prevents CV-V hiatus structures. Consequently, VHR results in a re-syllabification of hetero-syllabic consecutive vowel sequences, collapsing them into a single CV syllables, or separating them into adjacent CV.CV syllables.

Furthermore, the the nature of the VHR strategy employed is conditional upon two factors:

1) The features of the two vowels in hiatus combination.

2) The morpho-phonological boundary in which the hiatus sequence appears.

In other words, while the features of the vowels in hiatus sequences plays a role in determining the type of VHR applicable, the morpho-phonological context in which the hiatus sequence occurs presents additional constraints on the application of VHR, which override vowel-feature constraints, and lead to varying profiles of VHR in the three morpho-phonological contexts. To elaborate, vowel hiatus at morpheme boundaries is resolved using V1 elision, glide formation, and diphthongization based on the features of the two vowels involved. VHR at syllable boundaries employs only glide formation and diphthongization. The variation in VHR between the two contexts lies in the application of diphthongization. At syllable boundaries, any sequence involving a lower V1 and a higher V2 undergoes diphthongization. By contrast, at morpheme boundaries diphthongization is restricted to hiatus sequences involving the low V1 vowel /a/ in combination with any other higher V2 vowel. Here, close-mid vowels /e̞ and o̞/ undergo elision when followed by a high V2. At word boundaries, on the other hand, vowel hiatus is always resolved using glottal spot epenthesis, regardless of the sequences of vowels involved.

Cross-linguistic evidence suggests that numerous factors licence or block the occurrence of vowel hiatus, such as - "the nature of the prosodic or morpho-syntactic boundary at which hiatus arises, prominence of factors such as stress, vowel length and tone, minimal word length or weight conditions, the lexical or functional status of particular morphemes, rate of speech, and sensitivity to particular lexical items" (Casali, 2011). For example, vowel elision always targets V1 regardless of its position in the prefix or stem in chiShona; in ciCewa, however, it targets V1 in prefix positions but V2 in suffix positions (Casali, 1997; Kadenge and Simango, 2014). This study on Setswana, therefore, contributes information on typological variations found in the factors that license or block VHR.

Furthermore, it is also typologically very common for languages to employ more than one VHR strategy. Bantu languages, which generally do not allow hetero-syllabic vowel hiatus sequences, use multiple strategies to resolve them (Vratsanos and Kadenge, 2017). For instance, to repair hiatus, ciNsenga employs glide formation, elision, and secondary articulation (Kadenge and Simango, 2014); chiShona employs five strategies; glide formation, secondary articulation, elision, vowel coalescence, and glide epenthesis (Mudzingwa and Kadenge, 2014). Interestingly, hiatus resolution operates on different morpho-syntactic and phonological contexts in both these languages (Kadenge and Simango, 2014). Zezuru employs glide formation, glide epenthesis, vowel elision, and vowel coalescence to resolve hiatus (Kadenge, 2010). Both XitSonga (Vratsanos and Kadenge, 2017) and isiNdebele

(Sabao, 2015) use glide formation, secondary articulation, elision, and coalescence. Lastly, Simbiti uses five strategies, namely vowel elision, diphthongization, glide formation, and glide epenthesis (Overton, 2018). Evidence of multiple VHR strategies working in tandem found in Setswana contributes to the growing body of literature on VHR in other Bantu languages (ciNsenga and chiShona (Kadenge and Simango, 2014), (Bickmore, 2021), Simbiti (Overton, 2018) and Xitsonga (Vratsanos and Kadenge, 2017), Embosi (Rialland et al., 2015)), and provides an enriched understanding of factors that drive VHR in the language family.

This paper is structured as follows: Section 2 describes the methods of data collection and elicitation design. Section 3 provides preliminary information on Setswana's phonology, tone system, syllable structure, phonological processes, word classes, and word-formation processes. Section 4 consists of a detailed analysis of the different VHR strategies employed in the three different morpho-phonological contexts. This sections also provides a brief insight into VHR at stem-suffix interfaces (4.7). Finally, Section 5 presents the conclusion of the study.

2 Method

For the purpose of this project, 9:26 hours of audio data focusing on VHR in Setswana was elicited. An H4n-zoom recorder, provided by the Linguistics Department at the University of Rochester, was the primary equipment used for audio recordings. Acoustic analysis of speech utterances was performed using PRAAT on a MacOS. The speaker's vowel chart was created using *NORM: Vowel Normalization Suite* (Thomas and Kendall, 2007).

2.1 Elicitation design

All elicitations were designed using Creissels' (1996) extensive database of Setswana nouns and verbs and the online *Bab.la* Setswana dictionary (Bab). In every context, each V1 vowel was tested with examples of every other available V2 vowel. The database currently consists of 114 examples of VHR at morpheme boundaries, 33 examples of VHR at syllable boundaries, and 70 examples of VHR at word boundaries. Refer *Appendix* for all examples.

To test for hiatus resolution at morpheme boundaries, 60 nouns with vowel-initial stems were tested against vowel-final noun class prefixes *ma-, mo-, lɛ-, sɛ-* and *di-*.

Similarly, verbs consisting of varied vowel-initial stems such as ***à**pàjà* 'cook', ***á**dìmà* 'lend', ***à**là* 'make bed', ***é**mà* 'stand', ***ɛ̀**pà* 'dig', ***ɛ̀**tà* 'visit', ***ò**pɛ̀là* 'sing', ***í**kàɲà* 'depend', and ***ú**tɬwà* 'hear' were tested with the vowel-final subject marker prefixes *k**è**-* 'I', *r**ɛ̀**-* 'we', *b**á**-* 'they', *ò-* 'you, *ó-* 's/he', and *x**ò**-* the infinitive form.

Furthermore, to test for vowel hiatus resolution at word-boundaries, numerous words with varied final-vowels (*bɔ̀n**à*** 'see' and *rɛ̀kìl**ɛ̀*** 'bought', *mòt^h**ù*** 'person', *mòsìmàn**è*** 'boy', *mòsàd**ì*** 'woman' and personal names

Theo, Isago, and *Ubani*) were used in combination with nouns with varied initial-vowels (***à****pòlɛ̀* 'apple', ***à****ŋ̀ànsì* 'onion', ***ɛ̀****mɛ̀rɛ̀* 'bucket', ***ɛ̀****nʤɛ̀nɛ̀* 'engine', ***ɔ̀****vènè* 'oven', ***ɔ̀****mfɔ̀lɔ̀pɔ̀* 'envelope', ***ò****fìsì* 'office', and ***I****sago* and ***U****bani*). These words, including both nouns and verbs, were then used to design simple sentence constructions that display the different of V1-V2 hiatus combinations at verb-noun (5) and noun-noun (6) boundaries as shown below.

(5) *kè-bɔ̀n-***à** **à***pòlè*
1SG-see.PRS-FV 9.apple
'I see an apple.'

(6) *kè-f-á* *mò-sì̀mànɛ̀* **ɔ̀***mfɔ̀lɔ̀pɔ̀*
1SG-give.PRS-FV 1-boy 9.envelope
'I give the boy an envelope.'

Finally, 33 words (nouns, adjectives, and verbs) were selected from Criessel's (1996) database and *Bab.la* dictionary to test for vowel hiatus resolution at syllable boundaries.

During elicitation sessions, the speaker was asked to pronounce each example slowly, then at regular speed, tap out the number of syllables, and hum the tone.

To analyse how VHR affects tone, combinations of consecutive low vowels (L-L), low-high (L-H) vowels, high-low (H-L) vowels and consecutive high tone (H-H) vowels sequences were chosen. For instance, the high-tone subject agreement prefixes *bá-* 'they' and *ó-* 's/he', and the toneless subject agreement prefixes *kè-* 'I', *rɛ̀-* 'we', *ò-* 'you', and *xò-* were used in combination with various verb stems consisting of both high (*ádìmà* 'lend') and toneless initial vowels (*àpàjà* 'cook'). Similarly, the high-tone second noun-class prefix *bá-* was tested along with multiple noun stems consisting of both high and toneless initial vowels.

2.2 Consultant Profile

All language data for this project was collected in collaboration with a native Setswana speaker from Phitshane Molopo, Botswana. The speaker is currently an undergraduate Chemical Engineering student at the University of Rochester, USA, and also speaks fluent English.

3 Phonological and Grammatical Preliminaries

This section provides a brief overview of all preliminary information required for a detailed comprehension of this study. It includes information on Setswana's phoneme inventory 3.1, tone 3.2, phonological processes 3.3, syllable structure, word-classes, and word-formation processes 3.4.

3.1 Phoneme Inventory

The following is a description of the Setswana phoneme inventory based largely on data presented in Cole (1955) and Bennett et al. (2016), in addition to phonetic data elicited with the Botswanan speaker. Data collected during the project mostly converges with the information presented in Bennett et al. (2016). Minor variations are discussed below.

3.1.1 Consonants

Setswana, as spoken by the speaker, has 31 consonants as shown in Table 1

Table 1: **Consonant Inventory**

	Bilabial	**Labio-Dental**	**Alveolar**	**Alveolar Lateral**	**Post-Alveolar**	**Palatal**	**Velar**	**Uvular**	**Glottal**
Plosive	p p^h b		t t^h d				k k^h		(ʔ)
Affricate			ts ts^h	tɬ $tɬ^h$	ʧ $ʧ^h$ ʤ			qχ	
Nasal	m		n			ɳ	ŋ		
Trill			r						
Fricative		(f)	s z		ʃ			χ	h
Approximant	w			l		j			

Setswana's bilabial and alveolar plosives display a three-way contrast: voiceless unaspirated, voiceless aspirated, and voiced. Voiceless unaspirated plosives and affricates can be realized as ejectives (Bennett et al., 2016). /f/ is known to alternate with /h/ in several dialects, as seen in: [*khùtàfàtsà*] ⇄ [*khùtsàhàtsà*] 'make (oneself) sad.' Additionally, /b/ varies with /ts/, in some words such as [*màbɔ̀χɔ̀*] ⇄ [*màtsɔ̀χɔ̀*] 'arms'. The four nasal sounds /m/, /n/, /ɳ/, and /ŋ/ can be syllabic (Cole, 1955). Also, /d/, the voiced alveolar plosive, occurs only before the phonemic high vowels /i/ and /u/. It is listed as an allophone of /l/, because [l] never appears before /i/ or /u/ (Bennett et al., 2016).

A consonant word-list is given below in Table 2.

Table 2: **Consonant Word-list**

Phoneme	**Word (IPA)**	**Gloss**
/p/	pìnà	'song'
/t/	tàú	'lion'
/k/	kùbù	'hippopotamus'
/pʰ/	pʰìrì	'hyena'
/tʰ/	tʰàbà	'mountain'
/kʰ/	kʰùdù	'tortoise'
/b/	bìnà	'dance'
/d/	dìʤɔ́	'food'
[ʔ]	ʔáχá	'build'
/ts/	tsɛ̀bɛ́	'ear'
/ʧ/	sɛ̀ʧàbà	'nation'
/tɬ/	tɬálà	'famine'
/tsʰ/	tsʰɛ́χà	'laugh'
/ʧʰ/	ʧʰúswànì	'ant'
/tɬʰ/	tɬʰàpì	'fish'
/ʤ/	ʤá	'eat'
/qχ/	qχósì	'king'
/f/	fùlà	'graze'
/s/	sèhèlèlà	'cut'
/z/	zú	*'zoo'
/ʃ/	màʃì	'milk'
/χ/	χóχà	'pull'
/h/	hùmà	'become rich'
/m/	mòtʰù	'person'
/n/	nàmà	'meat'
/ɳ/	ɳàlà	*'marry'
/ŋ/	ŋòŋòrɛ̀χà	'complain'

Table 2: **Consonant Word-list**

Phoneme	**Word (IPA)**	**Gloss**
/r/	rútà	'teach'
/l/	lɛ́là	'cry'
/w/	wà	'fall'
/j/	jà	'go'

Occasionally, the variety of Setswana documented here displays certain variations from what is documented in the literature. For instance, while Cole (1955) lists the velar fricative /x/, Bennett et al. (2016) identify the sound as the uvular /χ/. Based on the speaker's assessment of the sound, the uvular fricative has been included in this consonant inventory. Also, the phoneme /z/ appears in the language as a result of language contact and borrowing, as seen in *zù* 'zoo.'; interestingly, this is documented in Cole (1955), but not in Bennett et al. (2016). Furthermore, although both Cole (1955) and Bennett et al., (2016) list *ʧúmà* 'hunt' as an example of the /ʧ/ phoneme, the speaker pronounces the word as *tsúmà* with a /ts/ instead. It was confirmed that the /ʧ/ sound does exist in the language, in words such as *sèʧàbà* 'nation.'

Lastly, although the glottal stop is non-phonemic, it can appear at the start of underlying vowel-initial stem, or between a vowel-initial stem and a vowel-final prefix (Bennett et al., 2016), as in *χɔ̀-ɛ́mà* → *χɔ̀-ʔɛ́mà* 'to stand'. This appears to be an instance of glottal stop epenthesis being used in VHR. However, in the provided context of vowel-final prefix and vowel-initial stem, none of the examples elicited in this documentation project yielded glottal stop epenthesis. This could be a variation between the Botswanan variety and South African variety of Setswana documented in Bennett et al. (2016). As explained in Section 4, VHR using glottal stop epenthesis is employed only at word-boundaries in the Botswanan variety. The given hiatus sequence would be repaired with a glide, as in *χɔ̀-ɛ́mà* → *χ^{w}ɛ́mà*.

3.1.2 Vowels

Setswana has 7 contrastive vowels - /i e̱ ɛ a ɔ o̱ u/ (Bennett et al., 2016), as shown in Figure 1. Converging with the data presented in Bennett et. al. (2016), the close-mid vowels are slightly higher than [e] and [o], and are therefore transcribed as raised close-mid vowels /e̱/ and /o̱/.

The mid vowels, both open-mid (/ɛ ɔ/) and close-mid (/e̱ o̱/) have raised allophones [ɛ̝ ɔ̝] and [i̱ u̱], respectively (Bennett et al., 2016). According to the literature (Bennett et al., 2016; Cole, 1955) this raising is conditioned by

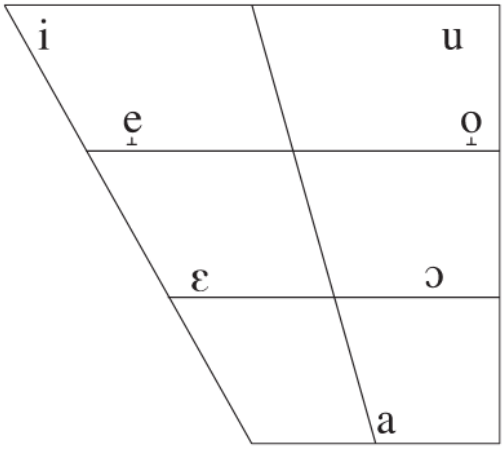


Figure 1: Vowel Chart, Bennett et al., 2016; pg 242

the presence of a high vowel in the succeeding syllable, the locative suffix [-ŋ], or the consonants /s, ʃ, ts, ts^{h}, ʧ, ʧ^{h}, tɬ, tɬ^{h}, ɲ/. For the project's primary speaker, this raising was significantly variable, appearing in certain contexts but not in others that warranted it. Similar to the observations recorded in Bennett et. al., (2016), the distinction between the raised allophones of the close-mid vowels /e̞ and o̞/ and the underlying high vowels /i and u/ is unclear for the speaker. There is also considerable overlap between the raised allophones of the open-mid vowels /ɛand ɔ/ and the close-mid vowels /e̞ and o̞/. A vowel word-list is given in Table 3

Table 3: **Vowel Word-list**

Phoneme	**Word (IPA)**	**Gloss**
/i/	ts^{h}ípì	'iron'
/e̞/	se̞χà	'cut'
/ɛ/	ts^{h}ɛ́χà	'laugh'
/a/	t^{h}àbà	'mountain'
/ɔ/	sɔ̀kà	'twist'
/o̞/	ʧo̞mà	'hunt'
/u/	ts^{h}úbà	'set alight'

A formant-scaled vowel plot of the speaker is given below in Figure 2[2] Measurements were taken at the midpoint of each vowel, averaged across three tokens for each example listed in Table 3.

[2] The vowels /ɛ/ and /ɔ/ are transcribed as /3/ and /0/ respectively, because NORM doesn't allow input of special IPA characters.

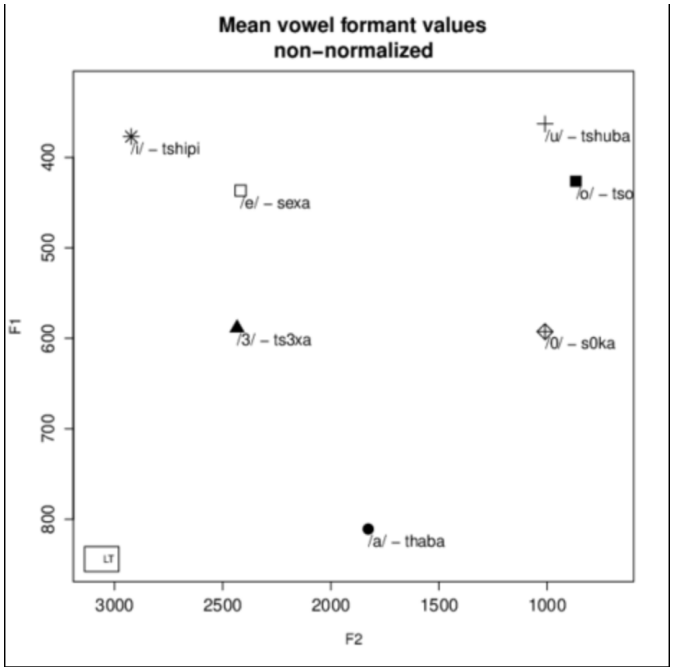


Figure 2: Formant-Scaled Vowel Plot

3.2 Tone

Setswana is a tonal language with a privative tone system. It distinguishes contrasts between marked (H)igh and inert (L)ow segments (Chebanne et al., 1997; Hyman, 2011; Yip, 2002). Additionally, Setswana exhibits progressive High Tone Spreading (HTS) in certain conditions. For instance, if an underlyingly L-toned syllable follows an H-toned syllable, it may take on a surface H-tone (Chebanne et al., 1997).

Evidence of HTS can be found in subject agreement marking on verb stems. (7) provides a clear tonal contrast between L-L and H-L constructions. Here, the only differentiating factor between the second person singular subject agreement marker *ò-* and the third person singular subject agreement marker *ó-* is tone. When prefixed to an underlyingly L-toned verb stem *-apaja* 'cook', the H-tone on *ó-* spreads, resulting in surface high tones across the verb stem, as seen in (7-b). By contrast, the L-toned *ò-* does not bring about any surface level changes on the verb stem.

(7) a. *wàpàjà nàmà*

ò-àpàjà nàmà
2SG-cook-FV 9.meat
'You cook meat.'

b. *wápájá nàmà*

ó-àpàjà nàmà
2SG-cook-FV 9.meat
'S/he cooks meat.'

This study examines tone in the restricted context of vowel hiatus and seeks to answer the question which naturally arises - what happens to the tone on the vowel segments, once VHR has applied? Since the tone bearing unit (TBU) is the syllable, vowels found in CV-V structures are marked for either H or L tone. This results in four possible tonal combinations; L-L, L-H, H-L, and H-H. As stated before, VHR is driven by a syllable structure constraint that results in the re-syllabification of hetero-syllabic vowels into a single vowel. Therefore, as a consequence of re-syllabification, it is observed that only one of the two tones marked on the adjacent vowels surfaces after VHR. Any hiatus sequence that features an H-tone, either underlyingly or due to processes of High Tone Spreading (HTS), surfaces with an H-tone after VHR.

Table 4 provides an overview of how tone surfaces after VHR.

Table 4: **Effects of VHR on Tone**

V1 Tone	**V2 Tone**	**Tone After VHR**	**Notes**
L	L	L	L being the inert, default tone, does not create any surface level changes.
L	H	H	Only the H-tone on V2 surfaces.
H	L	H	HTS causes surface H-tones on the nominal or verbal stem, resulting in an H-tone after VHR
H	H	H	Only the H tone on V2 surfaces.

3.3 Phonological Processes that Affect Vowels

Phonological processes are sensitive to morphological conditioning (Inkelas, 2014). They differ according to factors such as lexical class (noun or verb), morpheme type (stem or affix), and etymological origin (native words or loan words) (Sande, 2020). Cole (1955) describes the following four phonological processes that impact vowels:

Elision of vowels is observed in the noun class prefix mo-, The prefix's vowel /o/ is elided in contexts where it attaches to a noun stem beginning with /b/ or /f/. For example, **mo-b* → **mb-* → *mm- and *mo-f* → **mf-* → *mh-*. Additionally, in several dialects the vowels /e/, /ɛ/, and /o/ are elided when they are preceded and followed by

the liquid consonant /l/, resulting in the /l/ becoming syllabic. For example,

(8) a. *mólèlò* 'fire' → *móllò*

b. *sèhèlèlà* 'cut' → *sèhèllà*

Coalescence is the fusion of two juxtaposed sounds to produce a new intermediate one.

CV1 + V2 → .CV3(:).[3]

In Setswana, this occurs as follows: a + i > e, a + u > o, and a + a > a. However, Cole (1955) points out that the process is relatively uncommon in Setswana, and is mainly found in the formation of locatives with the suffix *-(i)ŋ*. An example of this can be found in (9).

(9) *thaba* 'mountain' + *(i)ŋ* → *thabeŋ* 'locative of mountain'

Consonantalization refers to the process by which a vowel changes into a semi-vowel with consonantal function. The front vowels /i, e̞, and ɛ/ change to [j], and the back vowels/u, o̞, and ɔ/ change to [w]. Cole (1955) notes that this process is mainly found in the formation of nominal and adjectival diminutives through suffixation of *-ana/-ane*. He also notes that the final vowel of the stem upon consonantalization further triggers a palatalization, velarization, or labialization of the preceding consonant. For example,

(10) a. *táù* 'lion' + *-ana* → *táwànà* 'lion cub'

b. *pódì* 'goat' + *-ane* → **pódìànè* → **pódyànè* → *pótsànè* 'baby goat.'

Vowel Alternation references an interchange of the back vowels /u, o̞, and ɔ/ with [w] plus a front vowel /i, e̞, ɛ, and a/, as shown below (adapted from Cole, (1955); pg 50).

(11) a. Close *u* interchanges with *wi*; *ntsu* ⇌ *ntswi* 'eagle'

b. Close-mid e̞ interchanges with *we̞*; *nole* ⇌ *nwele* 'perfect stem of *-nwa* 'drink'

c. Open-mid ɔ interchanges with *ɛ*; *lenχole* ⇌ *lenχwele* 'knee'

In Setswana, phonological processes such as nasalization, palatalization, velarization, and assimilation can also affect consonants (Cole, 1955). Interestingly, palatalization and velarization, in particular, become relevant in the analysis of VHR stem-suffix boundaries, where it is found that VHR through glide formation triggers a palatalization/velarization of the preceding consonant (refer 4.7).

[3] The schematic representation presented here, is based on Casali (1997). The parenthesized colon denotes optional compensatory lengthening. Additionally, periods are used to indicate syllable structure.

3.4 Word-formation Processes

Due to Setswana's agglutinative morphology, affixation is a very productive technique of word-formation (Krüger and Pretorius, 2006). Processes of prefixation and suffixation are used for both inflectional and derivational marking in the language (Cole, 1955). Affixes typically have a CV/V syllable structures. Coupled with the availability of vowel-initial stems, morpho-syntactic concatenation (both prefixation and suffixation) results in vowel hiatus sequences at several morpho-phonological boundaries (Kadenge and Simango, 2014).

3.4.1 Nouns

Setswana nouns consist of a nominal stem, which is preceded by a noun class prefix (1 to 14) which exhibit Bantu-typical systems of gender and agreement (Cole, 1955). As illustrated below, prefixation of the noun class morpheme to the nominal stem can result in hiatus sequences at the morpheme boundary.

(12) a. *mʷáχèdì̀*

mò-áχèdì̀
1-citizen
'citizen'

b. *sì̀.pà.tì̀*

sè-ì̀pàtì̀
7-excuse
'excuse'

Additionally, suffixes can be appended to the nominal stem to indicate augmentative or feminine (*-gadi/-adi*), diminutive (*-ana/-ane/-ɳana*), and locative (*-ŋ*) marking (Krüger and Pretorius, 2006). Vowel hiatus sequences are less common in stem-suffix structures. Nonetheless, they can be found in diminutive marking on nominal and adjectival forms. Diminutives have a variety of meanings, including small size or quantity, the young, or derogation and disparagement (Cole, 1955). Interestingly, unlike the majority of Bantu languages, Setswana uses suffixation for diminutive construction rather than prefixation, as shown below.

(13) *tá.wànà*

tá**ù-à**nà
lion-DIM
'lion cub.'

3.4.2 Verbs

The basic morphological structure of a Setswana verb requires either the infinitive morpheme *χo-* or agreement morphemes to be prefixed to the verb stem. Subject and object agreement marking, negation, and tense-aspect-mood inflections are also marked via prefixation of the morphemes to the verb stem (Krüger and Pretorius, 2006). Given below are some examples of hiatus sequences in infinitive marking (14) and subject agreement marking on a vowel-initial verb stem (15).

(14) *χú.tɬʷá*

χ**ò-ú**tɬʷ-á
15-hear-FV
'To hear.'

(15) a. *kʲà.pà.jà nà.mà*

k**è-à**pàjà nàmà
1SG-cook-FV 9.meat
'I cook meat.'

b. *ó.pɛ́.lá dì̀.pí́.nà*

ó-òpɛ̀l-à dì̀-pí́nà
3SG-sing-FV 8-song
'S/he sing songs.'

A rich variety of verbal extensions are also formed using the process of suffixation (Cole, 1955). These suffixes influence the verbs' lexical meaning and its syntactic combinability (Krüger and Pretorius, 2006). Setswana's most prominent verb extensions include passives (*-wa or -iwa*), neuters (*-exa, -ala, -esexa,* etc.), applicatives (*-ela, -etsa*), causatives (*-isa, -ya*), extensives (*-aka*), reciprocals (*-ana*), and reversives (*-ola*). Furthermore, a primitive verb stem usually has a terminative vowel *-a* (glossed FV). However, because this terminative vowel is substituted by various suffixal elements during conjugation (Cole, 1955; Krüger and Pretorius, 2006) vowel hiatus sequences at stem-suffix boundaries in verbal constructions are uncommon.

4 Analysis

This section analyzes the various VHR strategies used in Setswana. Each morpho-phonological context licences distinct applications of VHR. For example, morpheme boundaries warrant the usage of vowel elision, glide formation, and diphthongization. However, vowel sequences that elide at morpheme boundaries undergo diphthongization at syllable boundaries. Furthermore, vowel hiatus is always repaired using glottal stop epenthesis at word boundaries. The open-mid vowel /ɛ/ displays an interesting optionality in its choice of VHR strategy, undergoing either vowel elision or glide formation (4.3) at morpheme boundaries.

4.1 Vowel Elision

Vowel elision refers to the deletion of one of the two vowels, with or without compensatory lengthening.[4]

CV1 + V2 → .V1/2(:).

In Setswana, a mid V1 vowel /e̞ ɛ o̞ ɔ/ undergoes elision when it is followed by a homorganic high V2 vowel. There is no compensatory vowel lengthening.

Therefore, a front, close-mid vowel /e̞/ is elided when followed by the front, high vowel /i/. Similarly, the back, close-mid /o/ vowel undergoes elision when followed by a back, high vowel /u/.

$\mathbf{V_{MID} \rightarrow \emptyset \; / \; _ \; V_{HIGH}}$

$\mathbf{V_{FRONT\text{-}MID} \rightarrow \emptyset \; / \; _ \; V_{FRONT\text{-}HIGH}}$

$\mathbf{V_{BACK\text{-}MID} \rightarrow \emptyset \; / \; _ \; V_{BACK\text{-}HIGH}}$

Upon elision of V1, the tone on the V1 syllable also gets deleted. Therefore, L-L tone sequences, naturally, surface as L; L-H and H-H sequences surface as H; and, H-L sequences, as a result of HTS, also surface as H.

4.1.1 Morpheme Boundaries

Vowel elision occurs only at morpheme boundaries. The use of this strategy is not found in syllable or word boundaries. The following are examples of V1 elision at prefix-stem boundaries in noun class-noun stem (16) and subject agreement marker-verb stem (17) constructions.

[4] Although V1 elision is cross-linguistically far more common, attested in languages such as Yoruba, there are languages like Etsako that display V2 elision. (Casali, 1997).

(16) a. *sì.pà.tì*

sè-ìpàtì
7-excuse
‘excuse’

b. *mú.wà.nè*

mò-úwànè
3-mist
‘mist’

(17) a. *kí.kà.ɳà bò.nè*

kè-íkàɳ-à bònè
1SG-depend.on.PRS-FV them
‘I depend on them.’

b. *χú.tɬwà*

χò-útɬwà
INF-hear
‘To hear.’

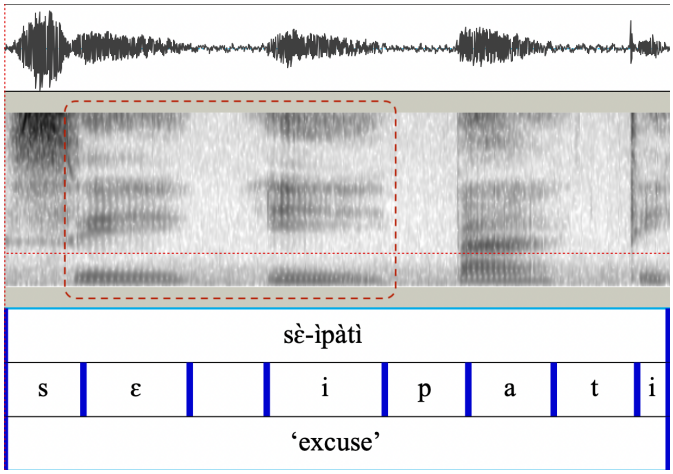


Figure 3: Morpheme Boundary - Before V1 Elision

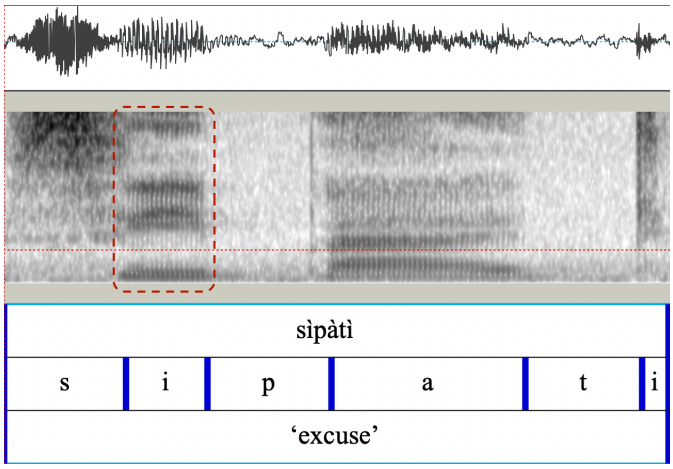


Figure 4: Morpheme Boundary - After V1 Elision

4.2 Glide Formation

Glide formation refers to the conversion of V1 into a semi vowel, as formalized below. [5]

CV1 + V2 → .CGV2(:).

In Setswana, vowel hiatus is repaired using glide formation in sequences where V1, a high or close-mid vowel /i u and e̱ o̱/, is followed by a homorganic lower vowel or a vowel with opposing backness. Front vowels convert into the palatal [j], while back vowels labialize into [w]. There is no compensatory lengthening.

$V_{HIGH/CLOSE\text{-}MID}$ → G / _ $V_{LOW/LOW\text{-}MID}$

i) $V_{FRONT,\ HIGH/CLOSE\text{-}MID}$ → j / _ $V_{FRONT,\ LOW/OPEN\text{-}MID;\ BACK}$

ii) $V_{BACK,\ HIGH/CLOSE\text{-}MID}$ → w / _ $V_{BACK,\ LOW/OPEN\text{-}MID;\ FRONT}$

Again, once V1 glides, the tone on the V1 syllable gets deleted, resulting in only one surface tone. Similar to V1 elision, L-L tone sequences, surface as L; L-H and H-H sequences surface as H; and, H-L sequences, as a result of HTS, also surface as H.

4.2.1 Morpheme Boundaries

The following are examples of glide formation at prefix-stem boundaries in noun class-noun stem (18) and subject agreement marker-verb stem (19) constructions.

(18)

a.	*dʲàpàrò*	b.	*mʷàxì*	c.	*mʲú.wà.nè*
	dì-àpàrò		mò-àxì		mè-úwànè
	8-cloth		1-Builder		4-mist
	‘clothes’		‘Builder’		‘mists’

[5] Cross-linguistically, languages can limit the availability of glide formation based on vowel backness and height. For instance, in Xhosa, only back round vowels glide, while in Polish only front vowels glide. Glide formation is more common when V1 is high, but Chicano Spanish has attested glide formation of mid V1s (Casali, 1995, 2011; Rosenthall, 1997). There are also languages like Ganda which displays glide formation of both high-front and high-back V1 vowels (Casali, 2011; Rosenthall, 1997).

(19) a. *kʲá.dì̀.mà bù.kà*

kè-ádì̀m-à bùkà
1SG-lend.PRS-FV 9.book
'I lend a book.'

b. *wɛ́.p.á m̀mù*

ó-ɛ̀p-à m̀mù
3SG-dig.PRS-FV 3.soil
'S/he digs soil.'

c. *kʲɔ̀.pè.là dì̀.pì̀.nà*

kè-ɔ̀pèl-à pì̀.nà
1SG-sing.PRS-FV 9.song
'I sing a songs.'

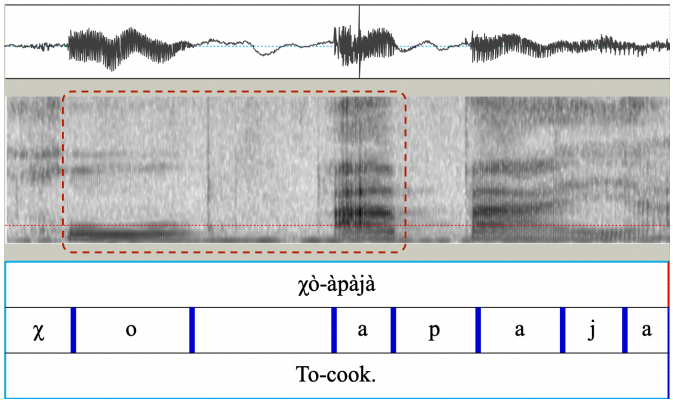


Figure 5: Morpheme Boundary - Before Glide Formation

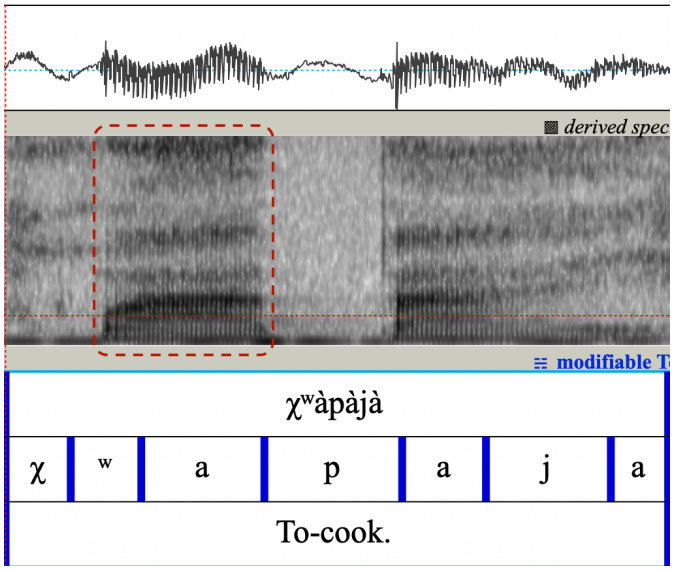


Figure 6: Morpheme Boundary - After Glide Formation

4.2.2 Syllable Boundaries

Glide formation is also found at word-internally at syllable boundaries.

(20) a. *mʲú.sʲá.mù*

mì.ú.sì.á.mù
9.museum
'museum'

b. *rá.dʲè.tà.rà*

rà.dì.e.tà.rá
9.radiator
'radiator'

c. *bʷì.pʷé.lè.tsò*

bù.ì.pù.é.lè.tsò
2a-repetition
'repetition'

d. *kù.kʷà.nà*

kù.kù.à.nà
9.gathering
'gathering'

4.3 The Case of /ɛ/

Interestingly, a open-mid V1 vowel /ɛ/ displays optionality in the choice of VHR strategy. It may either undergo vowel elision or glide formation when followed by the low vowel /a/ or back vowels /ɔ, o, and u/.

$V_{FRONT, OPEN\text{-}MID} \rightarrow$ Ø / _ $V_{LOW/BACK}$

OR

$V_{FRONT, OPEN\text{-}MID} \rightarrow$ G / _ $V_{LOW/BACK}$

This can be contrasted with the front close-mid vowel /e̯/, which elides when followed by a higher V2 (4.1), and glides when followed by a lower V2 (4.2).

Following are some examples of optional VHR displayed at prefix-stem boundaries, both noun class - noun stem, as seen in (21) and (22), and subject agreement marker - verb stem, as in (23) and (24). V1 elision is found in (21-a), (22-a), (23-a), and (24-a), while (21-b), (22-b), (23-b), and (24-b) reflect glide formation.

(21) a. *lɔ́bò*

lè-ɔ́bò
5-fence
'fence'

b. *lʲɔ́bò*

lè-ɔ́bò
5-fence
'fence'

(22) a. *sɔ́pò*

sè-ɔ́pò
7-group
'group'

b. *sʲɔ́pò*

sè-ɔ́pò
7-group
'group'

(23) a. *ràpàjà nàmà*

rè-àpàjà nàmà
1PL-cook.PRS-FV 9.meat
'We cook meat.'

b. *rʲàpàjà nàmà*

rè-àpàjà nàmà
1PL-cook.PRS-FV 9.meat
'We cook meat.'

(24) a. *rɔ̀lèlà ǹmù*

rè-ɔ̀lèlà ǹmù
1PL-pick.up.PRS-FV 3.dirt
'We pick up dirt.'

b. *rʲɔ̀lèlà ǹmù*

rè-ɔ̀lèlà ǹmù
1PL-pick.up.PRS-FV 3.dirt
'We pick up dirt.'

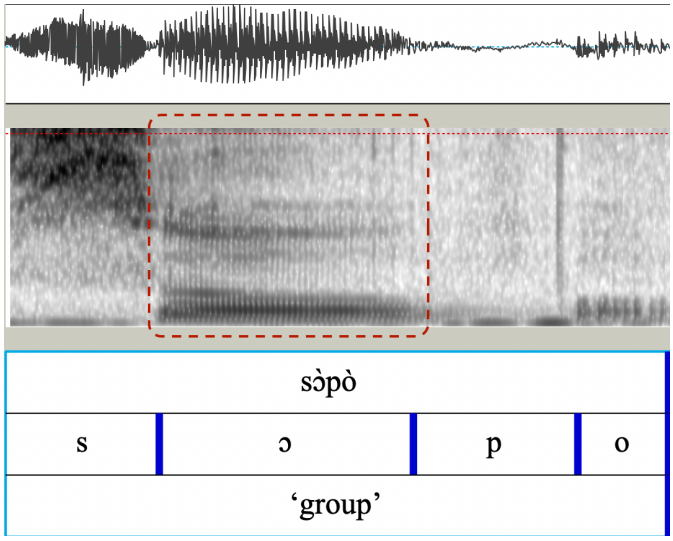


Figure 7: /ɛ/ - Resolved Using V1 Elision

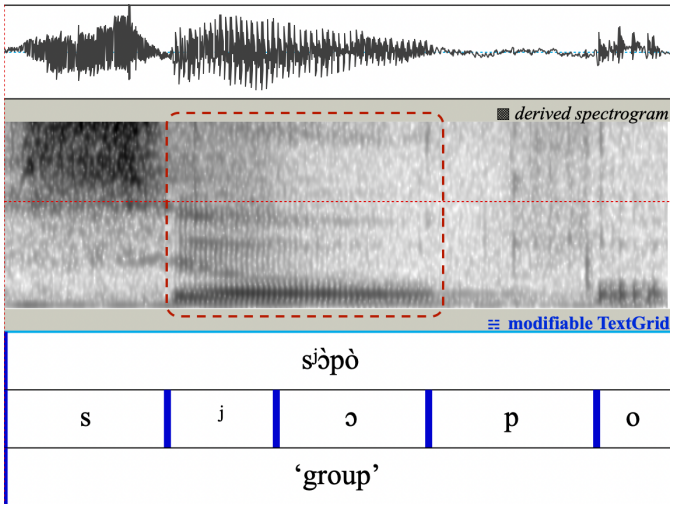


Figure 8: /ɛ/ - Resolved Using Glide Formation

Tone sequences are resolved similarly to vowel elision and glide formation.

4.4 Diphthongization

Diphthongization refers to the syllabification of the two vowels into the nucleus of a single syllable.

CV1 + V2 → .CV1V2.

The patterns of diphthongization appear to be slightly different at morpheme and syllable boundaries.

4.4.1 Morpheme Boundaries

Diphthongization occurs at morpheme boundaries exclusively when the hiatus sequence involves the low V1 vowel /a/ in combination with any higher V2 vowel (/i, u, ɛ, e̯, and o̯/), as shown below.

.V_{LOW}.$V_{HIGH/MID}$. → .$V_{LOW}V_{HIGH/MID}$.

Evidence of diphthongization can be found in the difference in how tone is marked on such constructions in slow and regular speech. In slow speech, hiatused vowels are marked with distinct tones, indicating that they heterosyllabic. In regular speech, by contrast, both vowels are marked with a single tone suggesting re-syllabification into one syllable. Converging with tonal patterns observed with vowel elision and glide formation, L-L sequences surface as L; and L-H, H-H and L-H sequences surface as H. Interestingly, diphthongs don't result in contour tones, at least as observed within the limited dataset of this project.

Below are some examples of diphthongization of hiatus sequences at prefix-stem boundaries in noun class-noun stem (16) and subject agreement marker-verb stem (26) constructions.

(25) a. *mái.kà.tɬò*

mà-í.kà.tɬò
14-regret
'regrets'

b. *bàɛ. mɪ̀.sɪ̀*

bà-ɛ̀mɪ̀sɪ̀
2-bridesmaid
'bridesmaids'

(26) a. *báo.pɛ́.lá dɪ̀.pɪ́.nà*

bá-ò.pɛ̀.l-à dɪ̀-pɪ́.nà
2PL-sing.PRS-FV 8-song
'They sing songs.'

b. *bái.ká.ŋá rò.nà*

bá.í.kà.ŋà rò.nà
2PL-depend.PRS-FV us
'They depend on us.'

4.4.2 Syllable Boundaries

On the other hand, at syllable boundaries, every hiatus sequence involving a lower V1 vowel and a higher V2 vowel undergoes diphthongization, as schematized below.

.$V_{LOW/MID}$.$V_{MID/HIGH}$. → .$V_{LOW/MID}V_{MID/HIGH}$.

Interestingly, vowel sequences consisting of close-mid V1 vowels (/e̯ and o̯/) and high V2 vowels /i and u/ that undergo elision 4.1 at morpheme boundaries are diphthongized at syllable boundaries as seen in (27-c) and (27-d).

(27) a. *m̀mài.lè*

m̀mà.ì.lè
9.mile
'mile'

b. *sè.kàu.sù*

sè-kà.ù.sù
7-sock
'sock'

c. *sóu.sò*

só.ù.sò
9.sauce
'sauce'

d. *t^{h}èi.pì*

t^{h}è.ì.pì
9.tape
'tape'

4.5 V1 = V2

In instances where the vowel hiatus sequence consists of two of the same vowels, the two hetero-syllabic vowels undergo re-syllabification into a single syllable. This is comparable to diphthongization, in which hetero-syllabic vowels are combined into a single syllable, except that the two vowels in this case are identical. In other words, the construction is pronounced as though only one were present. There is no compensatory lengthening.

Here again, only one tone surfaces after re-syllabification following the same patterns as observed above.

4.5.1 Morpheme Boundaries

Given below are examples of prefix-stem constructions with identical consecutive vowels, noun class - noun stem (28), and of subject agreement marker - verb stem constructions (29).

(28) a. *sè. lè*

sè-èlò
7-sentence
'sentence'

b. *má.nò*

mà-ánò
6-Plan
'plans'

(29) a. *xò.pè.là*

xò-òpèl-à
15-sing-FV
'To sing.'

b. *bá.dí.má pè.nè*

bá-ádìm-à pènè
2PL-borrow.PRS-FV 9.pen
'They borrow a pen'

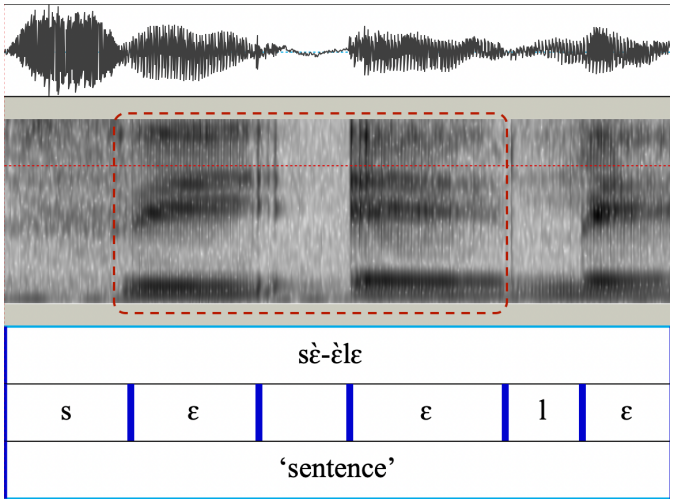


Figure 9: Morpheme Boundary with V1 = V2, Pronounced in Slow Speed

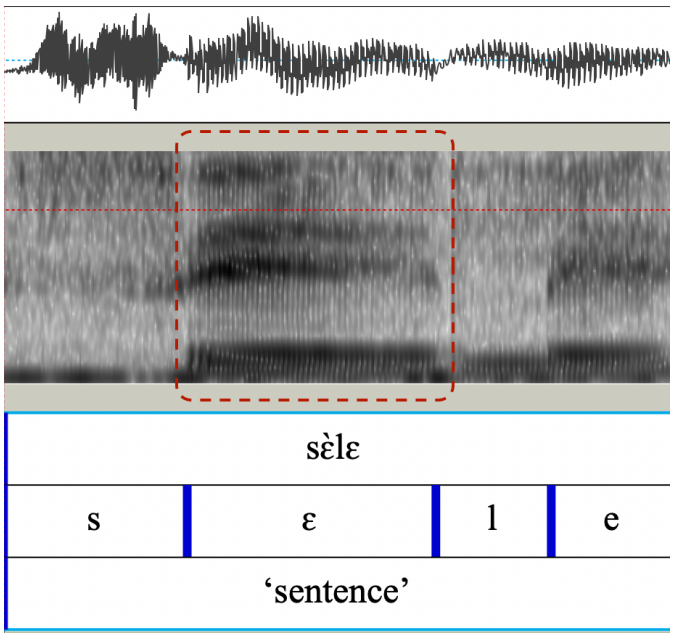


Figure 10: Morpheme Boundary with V1 = V2, Pronounced in Regular Speed

4.5.2 Syllable Boundaries

Re-syllabification of same vowel sequences into one syllable can also be found at syllable boundaries.

(30) a. *sà.tà.nì̀*

sà.à.tà.nì̀
9.satan
'satan'

b. *nɛ̀. lò*

nɛ̀. ɛ̀. lò
9.offer
'offer'

c. *lè.tɬò*

lè.tɬò.ò
5-hate
'hate'

d. *sè.kù.pù*

sè-kù.ù.pù
7-bottle
'bottle'

4.6 Glottal Stop Epenthesis

Epenthesis refers to the insertion of a glottal stop [ʔ] between V1 and V2 to interrupt hiatus. Epenthesis can also be achieved through the insertion of a homorganic glide that shares the same roundness as V1 or V2, a fricative [h], or a coronal consonant, such a rhotic or [t].

CV1 + V2 → CV1.CV2. or CV1.GV2. or CV1.ʔV2.

In Setswana, vowel hiatus is always repaired using glottal stop epenthesis at word boundaries, regardless of the sequence of vowels involved.

Ø → ʔ/ V# _ #V

In this paradigm, the two distinct tones on the vowel segments are retained as is.

4.6.1 Word Boundaries

Given below are a few examples showing vowel hiatus resolution at word boundaries between verbs-noun (31-a) (31-b) and noun-noun (31-c) (31-d) constructions.

(31) a. *kè-bɔ̀n-à^{ʔ}àpòlè*

kè-bɔ̀nà àpòlè
1SG-see.PRS-FV 9.apple
'I see an apple.'

b. *mò.t^{h}kè-rɛ́kɨ̀lɛ̀^{ʔ}ɔ̀vɛ̀nɛ̀*

kè-rɛ́kɨ̀lɛ̀ ɔ̀vɛ̀nɛ̀
1SG-buy-PST-FV 9.oven
'I bought an oven.'

c. *kè-fà mòsɨ̀mànè^{ʔ}ɛ̀mɛ̀rɛ̀*

kè-f-à mò-sɨ̀mànè ɛ̀mɛ̀rɛ̀
1SG-give.PRS-FV 1-boy 9.bucket
'I give the boy a bucket.'

d. *kè.fɨ̀lɛ̀ mòsàdɨ̀^{ʔ} ɔ̀mfɔ̀lɔ̀pɔ̀*

kè-fɨ̀l-ɛ̀ mò-sàdɨ̀ *ɔ̀mfɔ̀lɔ̀pɔ*
1SG-give-PST-FV 1-woman 9.envelope
'I give the woman an envelope.'

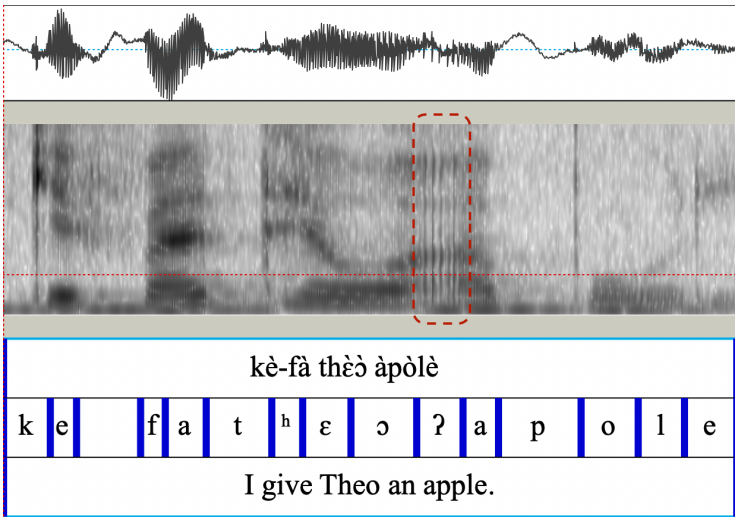


Figure 11: Glottal Stop Epenthesis

4.7 What happens at stem-suffix boundaries?

Due to the primarily substitutive nature of suffixation in Setswana (Krüger and Pretorius, 2006), vowel hiatus sequences are rare in stem-suffix contexts. However, limited examples of such sequences can be found in diminutive marking in the language, where the morpheme *-ana* suffixes to a noun or adjective stem. Therefore, the following analysis of VHR at stem-suffix boundaries is restricted to the interaction of noun stems consisting of varied final vowels with the diminutive suffix *-ana*, and therefore, focuses primarily on V1-a constructions.

VHR at stem-suffix boundaries appears to be different from the VHR at prefix-stem boundaries analysed above, at least in their surface realizations. In other words, despite the hiatus sequence comprising of the same combination of vowels, the resulting surface forms at prefix-stem and stem-suffix boundaries appear to be different, as shown below.

Hiatus sequences consisting of a high /i u/ or close-mid /e̯ o̯/ V1 vowel when followed by a low vowel tends to undergo glide formation as demonstrated in 4.2. However, diminutive suffixation results in the following surface forms.

(32) a. *kòlòɓʒʷànà*

kòlòbò-ànà
9.pig-DIM
'piglet'

b. *lòlèŋʷànà*

lòlèmè-ànà
9.tongue-DIM
'a small tongue'

(33) a. *tsʰìpìtsʷànà*

tsʰìpì-ànà
9.metal-DIM
'A small piece of metal.'

b. *lègòtʃʷànà*

lè-gòdù-ànà
5-thief-DIM
'A petty thief'

Instead of consistently being resolved using glide formation as seen prefix-stem constructions, the diminutives in (32) and (33) surface with *jʷa, ŋʷa, tsʷà, and tʃʷà*, exhibiting further palatalization/velarization in their forms. The surface forms that occur in stem-suffix boundaries are a result of an ordered sound change, as defined below.

1) VHR through glide formation.

2) Glide formation triggers a palatalization/velarization of the preceding consonant.

Evidence of such ordered sound changes has previously been documented in (Krüger and Pretorius, 2006) and (Cole, 1955). Palatalization and velarization are well-attested phonological process in Setswana. Cole (1955) defines the following rules for palatalization and velarization, which accurately correspond with the 'varied' structures seen above.

Rules of Palatalization/Velarization (Cole, 1955)

1. If the consonant in the final syllable of the noun is bilabial (p, b, pʰ, f), it undergoes palatalization; if the final consonant is the bilabial nasal m, it undergoes velarization.
b → **ɓʒʷ**; *kolobe* ('pig') → *koljʷane* ('piglet')

p → **tsʷ**; tshipi ('metal') → tshitsʷana ('small piece of iron')

ph → **tʃʷ**; marophi ('blisters') → marotʃʷana ('small blisters')

f → **tʃ^{w}**; gaufi (‘nearby’) → gautʃ^{w}ane (‘fairly nearby’)

m → **ŋ^{w}**; molomo (‘mouth’) → moloŋ^{w}ana (‘small mouth’)

2. If the consonant in the final syllable of the noun is an alveolar (t, r, d, l, n), preceding a high front vowel it undergoes palatalization; d also undergoes palatalization when followed by u. **t** → **ts**; lobati (‘plank’) → lobatswana (‘small plank’)

   r → **ts^{h}**; nare (‘buffalo’) → natshana (‘buffalo calf’)

   du → **ts^{w}**; legodu (‘thief’) → legotswana (‘little thief’)

   di→ **ts**; podi (‘goat’) → potsane (‘potsane’)

   n → **ɳ**; mosimane (‘boy’) → mosimaɳana (‘little boy’)

   l → **j**; mmele (’body’) → mmejana (‘little body’)

Palatalization and velarization in Setswana are a consequence of the incompatibility of bilabial, alveolar, velar, and glottal consonants with the glides -w and -j (Cole, 1955), and are found across many environments in the language. For instance, formation of passive verbs is achieved via suffixation of the morpheme *–wa* to the verb stem. This –w in the suffix leads to palatalization/velarization of the preceding consonants, as shown below:

(34) a. *-lopa* (‘request’) + *wa* → passive *–lotswa*

b. *-roba* (‘break’) + *wa* → passive *–rojwa*

c. *-roma* (‘send) + *wa* → passive *-rongwa*

The glide -w is crucial in triggering such palatalization/velarization in the above passive forms. Suffixation in diminutives would not demonstrate similar palatalization/velarization if there wasn’t an ordered sound change. VHR through glide formation had to occurr first, in order for the new glide to trigger a second step palatalization/velarization of the preceding consonant. The question arises as to why glide formation in prefix-stem environments does not trigger the palatalization of preceding consonants. The answer might lie in whether VHR and palatalization/velarization are two different types of phonological alternations. Cole (1955) notes that these rules obtain except when the final vowel of the noun is an /-a/, resulting in a-a sequences.

(35) a. *t^{h}ìpànà*

ts^{h}ìpà-ànà
9.knife-DIM
'A small knife.'

b. *t^{h}àbànà*

t^{h}àbà-ànà
9.mountain-DIM
'A small mountain.'

This paradigm seems to match with the analysis of V1=V2 vowels given in 4.5, where in sequences of identical vowel undergo re-syllabification into a single syllable.

5 Conclusion

In conclusion, Setswana, disallows vowel hiatus sequences, and utilizes various VHR strategies to repair hiatus sequences which occur in different morpho-phonological boundaries. VHR in the language occurs as a consequence of a syllable structure constraint that disallows long vowels, and therefore prevents CV-V type structures. VHR is a fast/regular speech phenomenon, which when applied results in the re-syllabification of hetero-syllabic adjacent vowels into one CV syllable or separate CV.CV syllables. VHR in Setswana is conditioned by the features of the vowels involved, but more importantly by the context in which the hiatus sequence occurs. That is, while the features of the vowels determine the type of VHR – vowel elision, glide formation, diphthongization, or glottal stop epenthesis – it is the morpho-phonological context that further licences or blocks the application VHR. This results in variations in VHR across the different boundaries. The following is a brief summary of VHR in Setswana:

1. Close-mid V1 vowels /e̯ and o̯/ undergo elision when succeeded by a high V2 vowel /i and u/ at morpheme boundaries.

2. High /i and u/ or close-mid (/e̯ and o̯/) V1 vowels, undergo glide formation when followed by any vowel lower V2 vowel. Front V1 vowels become palatal glides [j], and back V1 vowels become labial glides [w].

3. Hiatus sequences involving a lower V1 vowel and a higher V2 vowel undergo diphthongization into a one syllable at syllable boundaries. With a slight variation at morpheme boundaries, this strategy is confined to sequences involving the low V1 vowel /a/ followed by any higher V2 vowel. Hiatus sequences that elide at morpheme boundaries become diphthongs at syllable boundaries.

4. Hiatus sequences involving two identical hetero-syllabic vowels are re-syllabified into one syllable, without compensatory lengthening at morpheme and syllable boundaries. This converges with patterns of diphthongization, except that the vowel involved are identical.

5. The front, open-mid vowel /ɛ/ exhibits an optionality in VHR, undergoing either glide formation or vowel elision when followed by a low V2 /a/ or back vowels /ɔ, o̯, and u/.

6. At word boundaries, vowel hiatus is always resolved using glottal stop insertion, regardless of the vowel combination.

Being a tonal language, vowel hiatus produces CV-V structures with four possible tonal combinations, L-L, L-H, H-L, and H-H. VHR, thus, also impacts the tone on consecutive vowel segments. Since VHR causes the re-syllabification of hetero-syllabic vowels into mono-syllables, only one of the two tones surfaces after VHR. In brief, L-L tones surface as L; L-H and H-H tones surface as H; and H-L tones, as a result of HTS, surface as H. Interestingly, none of these sequences generate contour tones.

VHR at stem-suffix boundaries provides an interesting avenue for further research. Although palatalization and velarization are widely attested in the language, the palatal/velar forms that emerge seem rather unpredictable from the features of the consonants involved in the process. How or why these certain forms emerge seems, whether they are driven by phonetic or lexical processes requires further investigation. Additionally, vowel hiatus can occur between three consecutive vowels (Casali, 2011). This is also something that need to be studied and described.

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Appendix

A VHR at Morpheme Boundaries

Table 5: **VHR at Prefix-Stem Boundaries (SM-Verb Stem)**.

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
1.	a-a	bá-àpàjà nàmà	bápájá nàmà	They cook meat.	Re-syllabification	H-L	H (HTS)
2.	a-a	bá-ádìmà bùkà	bádímá bùkà	They lend a book.	Re-syllabification	H-H	H (HTS)
3.	a-ɛ	bá-ɛ̀tà xàè	báɛ́tá xàè	They visit home.	Diphthongization	H-L	H (HTS)
4.	a-ɛ	bá-ɛ̀pà m̀mù	báɛ́pá m̀mù	They dig soil.	Diphthongization	H-L	H (HTS)
5.	a-e	bá-émà fà	báemá fà	They stand here.	Diphthongization	H-H	H (HTS)
6.	a-ɔ	bá-ɔ̀pèlà dì-pìnà	báɔ́pélá dì-pìnà	They sing songs.	Diphthongization	H-L	H (HTS)

Table 5: **VHR at Prefix-Stem Boundaries (SM-Verb Stem)**.

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
7.	a-ɔ	bá-ɔ̀lèlà m̀mù	báɔ́lélá m̀mù	They pick up dirt.	Diphthongization	H-L	H (HTS)
8.	a-i	bá-íkàŋà ròná	báikáŋá ròná	They depend on us.	Diphthongization	H-H	H (HTS)
9.	a-u	bá-útɬwà módùmò	báutɬwá mɔ́dùmò	They hear noise.	Diphthongization	H-H	H (HTS)
10.	e-a	kè-àpàjà nàmà	kʲàpàjà nàmà	I cook meat.	Glide Formation	L-L	L
11.	e-a	kè-àlà bòlàò	kʲàlà bòlàò	I make the bed	Glide Formation	L-L	L
12.	e-ɛ	kè-ɛ̀tà bùsìχò	kʲɛ̀tà bùsìχò	I visit at night.	Glide Formation	L-L	L
13.	e-ɛ	kè-ɛ̀pà m̀mù	kʲɛ̀pà m̀mù	I dig soil.	Glide Formation	L-L	L
14.	e-e	kè-émà fà	kémà fà	I stand here.	Re-syllabification	L-H	H
15.	e-ɔ	kè-ɔ̀pèlà dì-pìnà	kʲɔ̀pèlà dì-pìnà	I sing songs.	Glide Formation	L-L	L
16.	e-ɔ	kè-ɔ̀lèlà m̀mù	kʲɔ̀lèlà m̀mù	I pick up dirt.	Glide Formation	L-L	L
17.	e-i	kè-íkàŋà bònè	kíkàŋà bònè	I depend on them.	V1 Elision	L-H	H
18.	e-u	kè-útɬwà módùmò	kʲútɬwà mɔ́dùmò	I hear noise.	Glide Formation	L-H	H
19.	ɛ-a ɛ-a	rɛ̀-àpàjà nàmà rɛ̀-àpàjà nàmà	rʲàpàjà nàmà ràpàjà nàmà	We cook meat.	Glide Formation V1 Elision	L-L L-L	L L
20.	ɛ-ɛ	rɛ̀-ɛ̀tà bùsìχò	rɛ̀tà bùsìχò	We visit at night.	Re-syllabification	L-L	L
21.	ɛ-ɛ	rɛ̀-ɛ̀pà m̀mù	rɛ̀pà m̀mù	We dig soil.	Re-syllabification	L-L	L
22.	ɛ-e	rɛ̀-émà fà	rémà fà	We stand here.	V1 Elision	L-H	H
23.	ɛ-ɔ ɛ-ɔ	rɛ̀-ɔ̀pèlà dì-pìnà rɛ̀-ɔ̀pèlà dì-pìnà	rʲɔ̀pèlà dì-pìnà rɔ̀pèlà dì-pìnà	We sing songs.	Glide Formation V1 Elision	L-L L-L	L L
24.	ɛ-ɔ	rɛ̀-ɔ̀lèlà m̀mù	rʲɔ̀lèlà m̀mù		Glide Formation	L-L	L

Table 5: **VHR at Prefix-Stem Boundaries (SM-Verb Stem)**.

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
	ɛ-ɔ	rɛ̀-ɔ̀lèlà m̀mù	rɔ̀lèlà m̀mù	We pick up dirt.	V1 Elision	L-L	L
25.	ɛ-i	rɛ̀-íkàŋà bònè	ríkàŋà bònè	We depend on them.	V1 Elision	L-H	H
26.	ɛ-u	rɛ̀-útɬwà módùmò	rútɬwà mɔ́dùmò	I hear noise.	V1 Elision	L-H	H
27.	o-a	χò-àpàjà	χ^{w}àpàjà	To cook.	Glide Formation	L-L	L
28.	o-a	χò-àlà	χ^{w}àlà	To make bed.	Glide Formation	L-L	L
29.	o-a	χò-ádìmà	χ^{w}ádìmà	To lend/borrow.	Glide Formation	L-H	H
30.	o-ɛ	χò-ɛ̀tà	χ^{w}ɛ̀tà	To visit.	Glide Formation	L-L	L
31.	o-ɛ	χò-ɛ̀pà	χ^{w}ɛ̀pà	To dig.	Glide Formation	L-L	L
32.	o-e	χò-émà	χ^{w}émà	To stand.	Glide Formation	L-H	H
33.	o-ɔ	χò-ɔ̀pèlà	χɔ̀pèlà	To sing.	Re-syllabification	L-L	L
34.	o-ɔ	χò-ɔ̀lèlà	χɔ̀pèlà	To pick up.	Re-syllabification	L-L	L
35.	o-i	χò-íkàŋà	χ^{w}íkàŋà	To depend on.	Re-syllabification	L-H	H
36.	o-u	χò-útɬwà	χútɬwà	To hear.	V1 Elision	L-H	H
37.	u-a	ù-àpàjà nàmà	wàpàjà nàmà	You cook meat.	Glide Formation	L-L	L
38.	u-a	ù-ádìmà bùkà	wádìmà bùkà	You lend a book.	Glide Formation	L-H	H
39.	u-ɛ	ù-ɛ̀tà bùsìxò	wɛ̀tà bùsìxò	You visit at night.	Glide Formation	L-L	L
40.	u-ɛ	ù-ɛ̀pà m̀mù	wɛ̀pà m̀mù	You dig soil.	Glide Formation	L-L	L
41.	u-e	ù-émà fà	wémà fà	You stand here.	Glide Formation	L-H	H
42.	u-ɔ	ù-ɔ̀pèlà dì-pìnà	ɔ̀pèlà dì-pìnà	You sing songs.	V1 Elision	L-L	L
43.	u-ɔ	ù-ɔ̀lèlà m̀mù	ɔ̀lèlà m̀mù	You pick up dirt.	V1 Elision	L-L	L

Table 5: **VHR at Prefix-Stem Boundaries (SM-Verb Stem)**.

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
44.	u-i	ù-íkàŋà bònè	wíkàŋà bònè	You depend on them.	Glide Formation	L-H	H
45.	u-u	ù-útɬwà módùmò	útɬwà módùmò	You hear noise.	Re-syllabification	L-H	H
46.	u-a	ú-àpàjà nàmà	wápájá nàmà	You cook meat.	Glide Formation	H-L	H (HTS)
47.	u-a	ú-ádìmà bùkà	wádímá bùkà	You lend a book.	Glide Formation	H-H	H (HTS)
48.	u-ɛ	ú-ɛ̀tà bùsìχò	wɛ́tá bùsìχò	You visit at night.	Glide Formation	H-L	H (HTS)
49.	u-ɛ	ú-ɛ̀pà m̀mù	wɛ́pá m̀mù	You dig soil.	Glide Formation	H-L	H (HTS)
50.	u-e	ú-émà fà	wémà fà	You stand here.	Glide Formation	H-H	H (HTS)
51.	u-ɔ	ú-ɔ́pèlà dì-pìnà	ɔ́pélá dì-pìnà	You sing songs.	V1 Elision	H-L	H (HTS)
52.	u-ɔ	ú-ɔ̀lèlà m̀mù	ɔ́lélá m̀mù	You pick up dirt.	V1 Elision	H-L	H (HTS)
53.	u-i	ú-íkàŋà bònè	wíkáŋá bònè	You depend on them.	Glide Formation	H-H	H (HTS)
54.	u-u	ú-útɬwà módùmò	útɬwá módùmò	You hear noise.	Re-syllabification	H-H	H (HTS)

Table 6: **VHR at Prefix-Stem Boundaries (NC-Noun Stem)**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
1.	a-a	mà-ànò	mànò	Plans	Re-syllabification	L-L	L
2.	a-a	mà-àróχì	màróχì	Rebels	Re-syllabification	L-L	L
3.	a-a	bá-áχì	báχì	Builders	Re-syllabification	H-H	H
4.	a-a	bá-áχèdì	báχèdì	Citizens	Re-syllabification	H-H	H
5.	a-ɛ	mà-ètò	màɛtò	Journeys	Diphthongization	L-L	L
6.	a-ɛ	mà-èbà	màɛbà	Doves	Diphthongization	L-L	L
7.	a-e	bá-ètsì	báetsì	Imitators	Diphthongization	H-L	H
8.	a-e	bá-èǹg	báeǹg	Visitors	Diphthongization	H-L	H
9.	a-e	bá-èmìsì	báemìsì	Bridesmaids	Diphthongization	H-L	H
10.	a-ɔ	mà-ɔ̀bò	màɔbò	Boundaries	Diphthongization	L-L	L
11.	a-ɔ	mà-ɔ́tò	máɔtò	Legs	Diphthongization	L-H	H
12.	a-o	bá-òkì	báokì	Nurses	Diphthongization	H-L	H
13.	a-o	bá-òkàmédì	báokàmédì	Supervisors	Diphthongization	H-L	H
14.	a-i	mà-ísò	máisò	Fireplaces	Diphthongization	L-H	H
15.	a-i	mà-íkàtɬò	máikàtɬò	Regrets	Diphthongization	L-H	H
16.	a-i	bá-ìtùdì	báitùdì	Students	Diphthongization	H-L	H
17.	a-u	mà-úbà	máubà	Droughts	Diphthongization	L-H	H
18.	ɛ-a	sɛ̀-ànè	sʲànè	Totem	Glide Formation	L-L	L
19.	ɛ-a	sɛ̀-àtɬà	sʲàtɬà	Hand	Glide Formation	L-L	L
20.	ɛ-a	sɛ̀-àpàrò	sʲàpàrò	Clothing	Glide Formation	L-L	L
21.	ɛ-a	lɛ̀-ànò	lʲànò	Plan	Glide Formation	L-L	L
22.	ɛ-a	lɛ̀-àróχì	lʲàróχì	Rebel	Glide Formation	L-H	H
23.	ɛ-ɛ	sɛ̀-ɛ̀lè	sɛ̀lè	Sentence	Re-syllabification	L-L	L
24.	ɛ-ɛ	sɛ̀-ɛ̀mò	sɛ̀mò	Situation	Re-syllabification	L-L	L
25.	ɛ-ɛ	lɛ̀-ɛ̀tò	lɛ̀tò	Journey	Re-syllabification	L-L	L
26.	ɛ-ɛ	lɛ̀-ɛ̀bà	lɛ̀bà	Dove	Re-syllabification	L-L	L

Table 6: **VHR at Prefix-Stem Boundaries (NC-Noun Stem)**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
27.	ɛ-ɔ	sɛ̀-ɔ́pò	sʲɔ́pò		Glide Formation	L-H	H
		sɛ̀-ɔ́pò	sɔ́pò	Group	V1 Elision	L-H	H
28.	ɛ-ɔ	sɛ̀-ɔ̀pèdì	sʲɔ̀pèdì		Glide Formation	L-L	L
		sɛ̀-ɔ̀pèdì	sɔ̀pèdì	Singer	V1 Elision	L-L	L
29.	ɛ-ɔ	lɛ̀-ɔ̀bò	lʲɔ̀bò		Glide Formation	L-L	L
		lɛ̀-ɔ́bò	lɔ́bò	Boundary	V1 Elision	L-H	H
30.	ɛ-ɔ	lɛ̀-ɔ́tò	lʲɔ́tò		Glide Formation	L-H	H
		lɛ̀-ɔ́tò	lɔ́tò	Leg	V1 Elision	L-H	H
31.	ɛ-i	sɛ̀-ìpátì	sìpátì	Excuse	V1 Elision	L-L	L
32.	ɛ-i	sɛ̀-ìpònè	sìpònè	Mirror	V1 Elision	L-L	L
33.	ɛ-i	lɛ̀-ísò	lísò	Fireplace	V1 Elision	L-H	H
34.	ɛ-i	lɛ̀-ìtɬò	lìtɬò	Eye	V1 Elision	L-L	L
35.	ɛ-u	lɛ̀-ú bà	lú bà	Drought	V1 Elision	L-H	H
36.	e-a	mè-àlò	mʲàlò	Coverings	Glide Formation	L-L	L
37.	e-a	mè-àχò	mʲàχò	Buildings	Glide Formation	L-L	L
38.	e-e	mè-èpò	mèpò	Mines	Re-syllabification	L-L	L
39.	e-i	mè-ìlà	mìlà	Taboos	V1 Elision	L-L	L
40.	e-u	mè-úwànè	mʲúwànè	Mists	Glide Formation	L-H	H
41.	o-a	mò-àlò	mʷàlò	Covering	Glide Formation	L-L	L
42.	o-a	mò-áxì	mʷáxì	Builder	Glide Formation	L-H	H
43.	o-a	mò-áχèdì	mʷáχèdì	Citizen	Glide Formation	L-H	H
44.	o-e	mò-èmìsì	mʷèmìsì	Bridesmaid	Glide Formation	L-L	L
45.	o-e	mò-èpò	mʷèpò	Mine	Glide Formation	L-L	L
46.	o-e	mò-èǹg	mʷèǹg	Visitor	Glide Formation	L-L	L
47.	o-e	mò-ètsì	mʷètsì	Imitator	Glide Formation	L-L	L
48.	o-o	mò-òkì	mòkì	Nurse	Re-syllabification	L-L	L

Table 6: **VHR at Prefix-Stem Boundaries (NC-Noun Stem)**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
49.	o-o	mò-òkàmédì	mòkàmédì	Supervisor	Re-syllabification	L-L	L
50.	o-i	mò-ìlà	m^{w}ìlà	Taboo	Glide Formation	L-L	L
51.	o-i	mò-ìtùdì	m^{w}ìtùdì	Student	Glide Formation	L-L	L
52.	o-u	mò-úwànè	múwànè	Mists	V1 Elision	L-H	H
53.	i-a	dì-ànè	d^{j}ànè	Totems	Glide Formation	L-L	L
54.	i-a	dì-àpàrò	d^{j}àpàrò	Clothes	Glide Formation	L-L	L
55.	i-a	dì-àtɬà	d^{j}àtɬà	Hands	Glide Formation	L-L	L
56.	i-e	dì-èlè	d^{j}èlè	Sentences	Glide Formation	L-L	L
57.	i-e	dì-èmò	d^{j}èmò	Situations	Glide Formation	L-L	L
58.	i-o	dì-òmìsì	d^{j}òmìsì	Driers	Glide Formation	L-L	L
59.	i-i	dì-ìpátì	dìpátì	Singers	Re-syllabification	L-L	L
60.	i-i	dì-ìpònè	dìpònè	Mirrors	Re-syllabification	L-L	L

B VHR at Syllable Boundaries

Table 7: **VHR at Syllable Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
1.	a-a	sà.à.tà.nì	sà.tà.nì	Satan	Re-syllabification	L-L	L
2.	a-ɛ	mú.là.ɛ̀.tsà	múlàɛtsà	Message	Diphthongization	L-L	L
3.	a-e	m̀mú.sà.è.sì	m̀mú.sàe.sì	Dictator	Diphthongization	L-L	L
4.	a-o	mò.là.ò	mò.lào	Law	Diphthongization	L-L	L
5.	a-o	là.ò.lè.là	lào.lè.là	Allow	Diphthongization	L-L	L
6.	a-i	m̀mà.ì.lè	m̀mài.lè	Mile	Diphthongization	L-L	L
7.	a-u	sè.mà.ù.sù	sè.màu.sù	Kiosk	Diphthongization	L-L	L

Table 7: **VHR at Syllable Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
8.	a-u	sì.kà.ù.sù	sì.kàu.sù	Socks	Diphthongization	L-L	L
9.	a-u	gà.ú.tà	gáu.tà	Gold	Diphthongization	L-H	H
10.	ɛ-ɛ	nɛ̀. ɛ̀. lò	nɛ̀. lò	Offer	Re-syllabification	L-L	L
11.	ɛ-ɔ	mò.t^{h}ɛ̀. ɔ̀	mò.t^{h}ɛɔ	Foundation	Diphthongization	L-L	L
12.	e-ɔ	pè.ɔ̀.mù.là.tò	p^{j}ɔ̀.mù.là.tò	Accusation	Glide Formation	L-L	L
13.	e-ɔ	tsè.ɔ̀.kà.rù.lò	ts^{j}ɔ̀.kà.rù.lò	Participation	Glide Formation	L-L	L
14.	e-i	t^{h}è.í.pì	t^{h}éi.pì	Tape	Diphthongization	L-H	H
15.	e-u	bà.bè.ù.dì	bà.b^{j}ù.dì	Barber	Glide Formation	L-L	L
16.	o-a	bò.bò.à.ŋà.nà	bò.b^{w}à.ŋà.nà	Fluffy	Glide Formation	L-L	L
17.	o-e	pò.è.lètsò	p^{w}è.lè.tsò	Repetition	Glide Formation	L-L	L
18.	o-i	bú.lò.ì	bú.lòi	Witchcraft	Diphthongization	L-L	L
19.	o-i	pò.ì.ts^{h}è.χò	pòi.ts^{h}è.χò	Phobia	Diphthongization	L-L	L
20.	o-u	só.ù.sò	só.ù.sò	Sauce	Diphthongization	H-L	H
21.	o-u	pò.ù.nà.mà	pòu.nà.mà	Lip	Diphthongization	L-L	L
22.	i-a	mì.ú.sì.á.mù	m^{j}ú.s^{j}á.mù	Museum	Glide Formation	L-H	H
23.	i-e	rà.dì.e.tà.rá	rá.d^{j}è.tà.rà	Radiator	Glide Formation	L-L	L
24.	i-o	dì.fì.o.sè	dì.f^{j}ò.sè	Fuse	Glide Formation	L-L	L
25.	i-i	tì.ì	tì	Tea	Re-syllabification	L-L	L
26.	i-i	bò.ì.dì.ì.dì	b^{w}ì.dì.dì	Multitudes	Glide Formation	L-L	L
27.	i-i	bò.ì.dì.ì.dì	b^{w}ì.dì.dì	Multitudes	Re-syllabification	L-L	L
28.	i-u	k^{h}ò.m.pì.ù.tà.rà	k^{h}ò.m.p^{j}ù.tà.rà	Computer	Glide Formation	L-L	L
29.	u-a	kù.kù.à.nà	kù.k^{w}à.nà	Gathering	Glide Formation	L-L	L
30.	u-e	bù.ì.pù.é.lè.tsò	b^{w}ì.p$_{w}$é.lè.tsò	Repetition	Glide Formation	L-H	H
31.	u-i	bù.ì.pù.é.lè.tsò	b^{w}ì.p$_{w}$é.lè.tsò	Repetition	Glide Formation	L-L	L
32.	u-i	pù.ì.sà.ŋó	p^{w}ì.sà.ŋó	Dialogue	Glide Formation	L-L	L
33.	u-u	sè.kù.ù.pù	sè.kù.pù	Bottle	Re-syllabification	L-L	L

C VHR at Word Boundaries

Table 8: **VHR at Word Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
1.	a-a	kè-bɔ̀nà àpɔ̀lè	kè-bɔ̀nàʔàpɔ̀lè	I see an apple.	Epentheis - Glottal Stop	L-L	L-L
2.	a-a	kè-bɔ̀nà àŋ̀ànsì	kè-bɔ̀nàʔàŋ̀ànsì	I see an onion.	Epenthesis - Glottal Stop	L-L	L-L
3.	a-ɛ	kè-bɔ̀nà ɛ̀mɛ̀rɛ̀	kè-bɔ̀nàʔɛ̀mɛ̀rɛ̀	I see a bucket.	Epenthesis - Glottal Stop	L-L	L-L
4.	a-ɛ	kè-bɔ̀nà ɛ̀nʤɛ̀nɛ̀	kè-bɔ̀nàʔɛ̀nʤɛ̀nɛ̀	I see an engine.	Epenthesis - Glottal Stop	L-L	L-L
5.	a-o	kè-bɔ̀nà ɔ̀fìsì	kè-bɔ̀nàʔɔ̀fìsì	I see an office.	Epenthesis - Glottal Stop	L-L	L-L
6.	a-ɔ	kè-bɔ̀nà ɔ̀vɛ̀nɛ̀	kè-bɔ̀nàʔɔ̀vɛ̀nɛ̀	I see an oven.	Epenthesis - Glottal Stop	L-L	L-L
7.	a-ɔ	kè-bɔ̀nà ɔ̀mfɔ̀lɔ̀pɔ	kè-bɔ̀nàʔɔ̀mfɔ̀lɔ̀pɔ̀	I see an envelope.	Epenthesis - Glottal Stop	L-L	L-L
8.	a-i	kè-bɔ̀nà ísàχò	kè-bɔ̀nàʔísàχò	I see Isago.	Epenthesis - Glottal Stop	L-H	L-H
9.	a-u	kè-bɔ̀nà úbànì	kè-bɔ̀nàʔúbànì	I see Ubani.	Epenthesis - Glottal Stop	L-L	L-L
10.	ɛ-a	kè-rɛ̀kìlɛ̀ àpɔ̀lè	kè-rɛ̀kìlɛ̀ʔàpɔ̀lè	I bought an apple.	Epentheis - Glottal Stop	L-L	L-L
11.	ɛ-a	kè-rɛ̀kìlɛ̀ àŋ̀ànsì	kè-rɛ̀kìlɛ̀ʔàŋ̀ànsì	I bought an onion.	Epenthesis - Glottal Stop	L-L	L-L
12.	ɛ-ɛ	kè-rɛ̀kìlɛ̀ ɛ̀mɛ̀rɛ̀	kè-rɛ̀kìlɛ̀ʔɛ̀mɛ̀rɛ̀	I bought a bucket.	Epenthesis - Glottal Stop	L-L	L-L

Table 8: **VHR at Word Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
13.	ɛ-ɛ	kè-rɛ̀kìlɛ̀ ɛ̀nʤɛ̀nɛ̀	kè-rɛ̀kìlɛ̀ʔɛ̀nʤɛ̀nɛ̀	I bought an engine.	Epenthesis - Glottal Stop	L-L	L-L
14.	ɛ-o	kè-rɛ̀kìlɛ̀ òfìsì	kè-rɛ̀kìlɛ̀ʔòfìsì	I bought an office.	Epenthesis - Glottal Stop	L-L	L-L
15.	ɛ-ɔ	kè-rɛ̀kìlɛ̀ ɔ̀vɛ̀nɛ̀	kè-rɛ̀kìlɛ̀ʔɔ̀vɛ̀nɛ̀	I bought an oven.	Epenthesis - Glottal Stop	L-L	L-L
16.	ɛ-ɔ	kè-rɛ̀kìlɛ̀ ɔ̀mfɔ̀lɔ̀pɔ	kè-rɛ̀kìlɛ̀ʔɔ̀mfɔ̀lɔ̀pɔ	I bought an envelope.	Epenthesis - Glottal Stop	L-L	L-L
17.	ɛ-i	kè-rɛ̀k`tsɛ̀ ísàχò pènè	kè-rɛ̀kɛ̀tsɛ̀ʔísàχò pènè	I bought Isago a pen.	Epenthesis - Glottal Stop	L-H	L-H
18.	ɛ-u	kè-rɛ̀kɛ̀tsɛ̀ úbànì pènè	kè-rɛ̀kɛ̀tsɛ̀ʔúbànì pènè	I bought Ubani a pen.	Epenthesis - Glottal Stop	L-L	L-L
19.	e-a	kè-fà mòsìmànè àpòlè	kè-fà mòsìmànèʔàpòlè	I give the boy an apple.	Epentheis - Glottal Stop	L-L	L-L
20.	e-a	kè-fà mòsìmànè àɳànsì	kè-fà mòsìmànèʔàɳànsì	I give the boy an onion.	Epenthesis - Glottal Stop	L-L	L-L
21.	e-ɛ	kè-fà mòsìmànè ɛ̀mɛ̀rɛ̀	kè-fà mòsìmànèʔɛ̀mɛ̀rɛ̀	I give the boy a bucket.	Epenthesis - Glottal Stop	L-L	L-L
22.	e-ɛ	kè-fà mòsìmànè ɛ̀nʤɛ̀nɛ̀	kè-fà mòsìmànèʔɛ̀nʤɛ̀nɛ̀	I give the boy an engine.	Epenthesis - Glottal Stop	L-L	L-L
23.	e-o	kè-fà mòsìmànè òfìsì	kè-fà mòsìmànèʔòfìsì	I give the boy an office.	Epenthesis - Glottal Stop	L-L	L-L
24.	e-ɔ	kè-fà mòsìmànè ɔ̀vɛ̀nɛ̀	kè-fà mòsìmànèʔɔ̀vɛ̀nɛ̀	I give the boy an oven.	Epenthesis - Glottal Stop	L-L	L-L
25.	e-ɔ	kè-fà mòsìmànè ɔ̀mfɔ̀lɔ̀pɔ	kè-fà mòsìmànèʔ ɔ̀mfɔ̀lɔ̀pɔ̀	I give the boy an envelope.	Epenthesis - Glottal Stop	L-L	L-L

Table 8: **VHR at Word Boundaries**

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
26.	e-i	kè-fà mòsìmànè ísàχò	kè-fà mòsìmànèʔísàχò	I give the boy Isago.	Epenthesis - Glottal Stop	L-H	L-H
27.	e-u	kè-fà mòsìmànè úbànì	kè-fà mòsìmànèʔúbànì	I give the boy Ubani.	Epenthesis - Glottal Stop	L-L	L-L
28.	e-ɛ	rè-ádìmà mòsìmànè èmɛ̀rɛ̀	rádìmà mòsìmànèʔèmɛ̀rɛ̀	We lend the boy a bucket.	Epenthesis - Glottal Stop	L-L	L-L
29.	e-ɔ	ú-ádìmà mòsìmànè ɔ̀vɛ̀nɛ̀	wádímá mòsìmànèʔɔ̀vɛ̀nɛ̀	S/he lends the boy a bucket.	Epenthesis - Glottal Stop	L-L	L-L
30.	ɔ-a	kè-fà tʰèɔ̀ àpòlè	kè-fà tʰèɔ̀ʔàpòlè	I give Theo an apple.	Epentheis - Glottal Stop	L-L	L-L
31.	ɔ-a	kè-fà tʰèɔ̀ àŋ̀ànsì	kè-fà tʰèɔ̀ʔàŋ̀ànsì	I give Theo an onion.	Epenthesis - Glottal Stop	L-L	L-L
32.	ɔ-ɛ	kè-fà tʰèɔ̀ èmɛ̀rɛ̀	kè-fà tʰèɔ̀ʔèmɛ̀rɛ̀	I give Theo a bucket.	Epenthesis - Glottal Stop	L-L	L-L
33.	ɔ-ɛ	kè-fà tʰèɔ̀ ènɡ̀ènè	kè-fà tʰèɔ̀ʔènɡ̀ènè	I give Theo an engine.	Epenthesis - Glottal Stop	L-L	L-L
34.	ɔ-o	kè-fà tʰèɔ̀ òfìsì	kè-fà tʰèɔ̀ʔòfìsì	I give Theo an office.	Epenthesis - Glottal Stop	L-L	L-L
35.	ɔ-ɔ	kè-fà tʰèɔ̀ɔ̀vɛ̀nɛ̀	kè-fà tʰèɔ̀ʔɔ̀vɛ̀nɛ̀	I give Theo an oven.	Epenthesis - Glottal Stop	L-L	L-L
36.	ɔ-ɔ	kè-fà tʰèɔ̀ ɔ̀mfɔ̀lɔ̀pɔ	kè-fà tʰèɔ̀ʔ ɔ̀mfɔ̀lɔ̀pɔ̀	I give Theo an envelope.	Epenthesis - Glottal Stop	L-L	L-L
37.	ɔ-i	kè-fà tʰèɔ̀ ísàχò	kè-fà tʰèɔ̀ʔísàχò	I give Theo Isago.	Epenthesis - Glottal Stop	L-H	L-H

Table 8: **VHR at Word Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
38.	ɔ-u	kè-fà tʰèɔ́ úbànì	kè-fà tʰèɔ́ʔúbànì	I give Theo Ubani.	Epenthesis - Glottal Stop	L-L	L-L
39.	ɔ-ɛ	rè-ádìmà tʰèɔ́ èmèrè	ràdìmà tʰèɔ́ʔèmèrè	We lend Theo a bucket.	Epenthesis - Glottal Stop	L-L	L-L
40.	ɔ-ɔ	ú-ádìmà tʰèɔ́ ɔ̀vènè	wádímá tʰèɔ́ʔɔ̀vènè	S/he lends Theo a bucket.	Epenthesis - Glottal Stop	H-L	H-L
41.	o-a	kè-fà ísàxò àpòlè	kè-fà ísàxòʔàpòlè	I give Isago an apple.	Epentheis - Glottal Stop	L-L	L-L
42.	o-a	kè-fà ísàχò àɳànsì	kè-fà ísàχòʔàɳànsì	I give Isago an onion.	Epenthesis - Glottal Stop	L-L	L-L
43.	o-ɛ	kè-fà ísàχò èmèrè	kè-fà ísàχòʔèmèrè	I give Isago a bucket.	Epenthesis - Glottal Stop	L-L	L-L
44.	o-ɛ	kè-fà ísàχò ènʤènè	kè-fà ísàχòʔènʤènè	I give Isago an engine.	Epenthesis - Glottal Stop	L-L	L-L
45.	o-o	kè-fà ísàχò òfìsì	kè-fà ísàχòʔòfìsì	I give Isago an office.	Epenthesis - Glottal Stop	L-L	L-L
46.	o-ɔ	kè-fà ísàχò ɔ̀vènè	kè-fà ísàχòʔɔ̀vènè	I give Isago an oven.	Epenthesis - Glottal Stop	L-L	L-L
47.	o-ɔ	kè-fà ísàχò ɔ̀mfɔ̀lɔ̀pɔ	kè-fà ísàχòʔ ɔ̀mfɔ̀lɔ̀pɔ̀	I give Isago an envelope.	Epenthesis - Glottal Stop	L-L	L-L
48.	o-i	kè-fà ísàχò ísàχɔ̀	kè-fà ísàχòʔísàχɔ̀	I give Isago Isago.	Epenthesis - Glottal Stop	L-H	L-H
49.	o-u	kè-fà ísàχò úbànì	kè-fà ísàχòʔúbànì	I give Isago Ubani.	Epenthesis - Glottal Stop	L-L	L-L
50.	o-ɛ	rè-ádìmà ísàχò èmèrè	ràdìmà ísàχòʔèmèrè	We lend Isago a bucket.	Epenthesis - Glottal Stop	L-L	L-L

Table 8: **VHR at Word Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
51.	o-ɔ	ú-ádìmà ísàχò ɔ̀vɛ̀nɛ̀	wádímá ísàχɔ̀ʔɔ̀vɛ̀nɛ̀	S/he lends Isago a bucket.	Epenthesis - Glottal Stop	H-L	H-L
52.	i-a	kè-fílɛ̀ mòsàdì àpòlè	kè-fílɛ̀ mòsàdìʔàpòlè	I give the woman an apple.	Epentheis - Glottal Stop	L-L	L-L
53.	i-a	kè-fílɛ̀ mòsàdì àɳànsì	kè-fílɛ̀ mòsàdìʔàɳànsì	I give the woman an onion.	Epenthesis - Glottal Stop	L-L	L-L
54.	i-ɛ	kè-fílɛ̀ mòsàdì èmèrɛ̀	kè-fílɛ̀ mòsàdìʔèmèrɛ̀	I give the woman a bucket.	Epenthesis - Glottal Stop	L-L	L-L
55.	i-ɛ	kè-fílɛ̀ mòsàdì ènʤènɛ̀	kè-fílɛ̀ mòsàdìʔènʤènɛ̀	I give the woman an engine.	Epenthesis - Glottal Stop	L-L	L-L
56.	i-o	kè-fílɛ̀ mòsàdì òfìsì	kè-fílɛ̀ mòsàdìʔòfìsì	I give the woman an office.	Epenthesis - Glottal Stop	L-L	L-L
57.	i-ɔ	kè-fílɛ̀ mòsàdì ɔ̀vɛ̀nɛ̀	kè-fílɛ̀mòsàdìʔɔ̀vɛ̀nɛ̀	I give the woman an oven.	Epenthesis - Glottal Stop	L-L	L-L
58.	i-ɔ	kè-fílɛ̀ mòsàdì ɔ̀mfɔ̀lɔ̀pɔ	kè-fílɛ̀ mòsàdìʔ ɔ̀mfɔ̀lɔ̀pɔ̀	I give the woman an envelope.	Epenthesis - Glottal Stop	L-L	L-L
59.	i-i	kè-fílɛ̀ mòsàdì ísàχɔ̀	kè-fílɛ̀ mòsàdìʔísàχɔ̀	I give the woman Isago.	Epenthesis - Glottal Stop	L-H	L-H
60.	i-u	kè-fílɛ̀ mòsàdì úbànì	kè-fílɛ̀mòsàdìʔúbànì	I give the woman Ubani.	Epenthesis - Glottal Stop	L-L	L-L
61.	u-a	kè-fílɛ̀ mòthù àpòlè	kè-fílɛ̀ mòthùʔàpòlè	I give the person an apple.	Epentheis - Glottal Stop	L-L	L-L
62.	u-a	kè-fílɛ̀ mòthù àɳànsì	kè-fílɛ̀ mòthùʔàɳànsì	I give the person an onion.	Epenthesis - Glottal Stop	L-L	L-L
63.	u-ɛ	kè-fílɛ̀ mòthù èmèrɛ̀	kè-fílɛ̀ mòthùʔèmèrɛ̀	I give the person a bucket.	Epenthesis - Glottal Stop	L-L	L-L

Table 8: **VHR at Word Boundaries**

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
64.	u-ɛ	kè-fìlɛ̀ mòthù ɛ̀nʤɛ̀nɛ̀	kè-fìlɛ̀ mòthùʔɛ̀nʤɛ̀nɛ̀	I give the person an engine.	Epenthesis - Glottal Stop	L-L	L-L
65.	u-o	kè-fìlɛ̀ mòthù òfìsì	kè-fìlɛ̀ mòthùʔòfìsì	I give the person an office.	Epenthesis - Glottal Stop	L-L	L-L
66.	u-ɔ	kè-fìlɛ̀ mòthù ɔ̀vɛ̀nɛ̀	kè-fìlɛ̀mòthùʔɔ̀vɛ̀nɛ̀	I give the person an oven.	Epenthesis - Glottal Stop	L-L	L-L
67.	u-ɔ	kè-fìlɛ̀ mòthù ɔ̀mfɔ̀lɔ̀pɔ	kè-fìlɛ̀ mòsthùʔ ɔ̀mfɔ̀lɔ̀pɔ̀	I give the person an envelope.	Epenthesis - Glottal Stop	L-L	L-L
68.	u-i	kè-fìlɛ̀ mòthù ísàχɔ̀	kè-fìlɛ̀ mòthùʔísàχɔ̀	I give the person Isago.	Epenthesis - Glottal Stop	L-H	L-H
69.	u-u	kè-fìlɛ̀ mòthù úbànì	kè-fìlɛ̀mòthùʔúbànì	I give the person Ubani.	Epenthesis - Glottal Stop	L-L	L-L

D Tone Analysis

Table 9: **Tone Analysis in Verbs**

S.no	**Underlying Form**	**Surface Form**	**Translation**	**V1-V2 Tone**	**Surface Tone**
1.	χò-àpàjà	χ^{w}àpàjà	To cook.	L-L	L
2.	kè-àpàjà nàmà	k^{j}àpàjà nàmà	I cook meat.	L-L	L
3.	rɛ̀-àpàjà nàmà	r^{j}àpàjà nàmà ràpàjà nàmà	We cook meat.	L-L	L
4.	bá-àpàjà nàmà	bápájá nàmà	They cook meat.	H-L	H (HTS)
5.	ù-àpàjà nàmà	wàpàjà nàmà	You cook meat.	L-L	L
6.	ú-àpàjà nàmà	wápájá nàmà	S/he cooks meat.	H-L	H (HTS)

Table 10: **Tone Analysis in Nouns**

S.no	**Underlying Form**	**Surface Form**	**Translation**	**V1-V2 Tone**	**Surface Tone**
1.	mà-ànò	mànò	Plans	L-L	L
2.	bá-áχì	báχì	Builders	H-H	H
3.	mà-ú bà	mábà	Droughts	L-H	H
4.	mò-áxì	m^{w}áxì	Builder	L-H	H
5.	bá-èmìsì	báemìsì	Bridesmaids	H-L	H
7.	bá-ètsì	báetsì	Imitators	H-L	H